

Traininig

April 29, 2025

```
[1]: # --- Install All in One Go ---
```

```
%pip install numpy tensorflow scikit-learn matplotlib pandas
```

```
Requirement already satisfied: numpy in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages
(2.1.3)
Requirement already satisfied: tensorflow in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages
(2.19.0)
Requirement already satisfied: scikit-learn in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages
(1.6.1)
Requirement already satisfied: matplotlib in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages
(3.10.1)
Requirement already satisfied: pandas in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages
(2.2.3)
Collecting
protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3
  Using cached protobuf-5.29.4-cp310-abi3-win_amd64.whl (434 kB)
Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (0.31.0)
Requirement already satisfied: termcolor>=1.1.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (3.0.1)
Requirement already satisfied: libclang>=13.0.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (3.4.0)
Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (0.5.1)
Requirement already satisfied: flatbuffers>=24.3.25 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
```

tensorflow) (25.2.10)
Requirement already satisfied: absl-py>=1.0.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (2.2.2)
Requirement already satisfied: six>=1.12.0 in
c:\users\reetam\appdata\roaming\python\python310\site-packages (from tensorflow)
(1.17.0)
Requirement already satisfied: h5py>=3.11.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (3.13.0)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (0.6.0)
Requirement already satisfied: setuptools in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (65.5.0)
Requirement already satisfied: google-pasta>=0.1.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (0.2.0)
Requirement already satisfied: astunparse>=1.6.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (1.6.3)
Requirement already satisfied: keras>=3.5.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (3.9.2)
Requirement already satisfied: typing-extensions>=3.6.6 in
c:\users\reetam\appdata\roaming\python\python310\site-packages (from tensorflow)
(4.13.2)
Requirement already satisfied: tensorboard~=2.19.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (2.19.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (1.71.0)
Requirement already satisfied: requests<3,>=2.21.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (2.32.3)
Requirement already satisfied: wrapt>=1.11.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorflow) (1.17.2)
Requirement already satisfied: packaging in
c:\users\reetam\appdata\roaming\python\python310\site-packages (from tensorflow)
(25.0)
Requirement already satisfied: scipy>=1.6.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
scikit-learn) (1.15.2)
Requirement already satisfied: joblib>=1.2.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from

```

scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
scikit-learn) (3.6.0)
Requirement already satisfied: pillow>=8 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (11.2.1)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (3.2.3)
Requirement already satisfied: cycler>=0.10 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (0.12.1)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (1.3.2)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (4.57.0)
Requirement already satisfied: python-dateutil>=2.7 in
c:\users\reetam\appdata\roaming\python\python310\site-packages (from matplotlib)
(2.9.0.post0)
Requirement already satisfied: kiwisolver>=1.3.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
matplotlib) (1.4.8)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
pandas) (2025.2)
Requirement already satisfied: pytz>=2020.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
pandas) (2025.2)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
astunparse>=1.6.0->tensorflow) (0.45.1)
Requirement already satisfied: rich in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
keras>=3.5.0->tensorflow) (14.0.0)
Requirement already satisfied: optree in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
keras>=3.5.0->tensorflow) (0.15.0)
Requirement already satisfied: namex in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
keras>=3.5.0->tensorflow) (0.0.9)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (2025.4.26)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from

```

```

requests<3,>=2.21.0->tensorflow) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (2.4.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (3.4.1)
Collecting tensorboard-data-server<0.8.0,>=0.7.0
  Using cached tensorboard_data_server-0.7.2-py3-none-any.whl (2.4 kB)
Requirement already satisfied: markdown>=2.6.8 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorboard~=2.19.0->tensorflow) (3.8)
Requirement already satisfied: werkzeug>=1.0.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
tensorboard~=2.19.0->tensorflow) (3.1.3)
Requirement already satisfied: MarkupSafe>=2.1.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
werkzeug>=1.0.1->tensorboard~=2.19.0->tensorflow) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
rich->keras>=3.5.0->tensorflow) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
c:\users\reetam\appdata\roaming\python\python310\site-packages (from
rich->keras>=3.5.0->tensorflow) (2.19.1)
Requirement already satisfied: mdurl~=0.1 in
c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages (from
markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow) (0.1.2)
Installing collected packages: tensorboard-data-server, protobuf
  Attempting uninstall: tensorboard-data-server
    Found existing installation: tensorboard-data-server 0.6.1
    Uninstalling tensorboard-data-server-0.6.1:
      Successfully uninstalled tensorboard-data-server-0.6.1
  Attempting uninstall: protobuf
    Found existing installation: protobuf 3.19.6
    Uninstalling protobuf-3.19.6:
      Successfully uninstalled protobuf-3.19.6
Successfully installed protobuf-5.29.4 tensorboard-data-server-0.7.2
Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)

```

```
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
WARNING: Ignoring invalid distribution -rotobuf
(c:\users\reetam\appdata\local\programs\python\python310\lib\site-packages)
```

[notice] A new release of pip is available: 23.0.1 -> 25.1

[notice] To update, run: python.exe -m pip install --upgrade pip

```
[23]: import csv

import numpy as np
import tensorflow as tf
from sklearn.model_selection import train_test_split

RANDOM_SEED = 42
```

```
[24]: dataset = 'keypoint_classifier/keypoint.csv'
# dataset = '/kaggle/input/mediapipe4/keypoint.csv'
model_save_path = 'keypoint_classifier.keras'
```

```
[25]: NUM_CLASSES = 26
```

```
[26]: X_dataset = np.loadtxt(dataset, delimiter=',', dtype='float32',
    ↪ usecols=list(range(1, (21 * 2) + 1)))
```

```
[27]: y_dataset = np.loadtxt(dataset, delimiter=',', dtype='int32', usecols=(0))
```

```
[28]: X_train, X_test, y_train, y_test = train_test_split(X_dataset, y_dataset,
    ↪ train_size=0.75, random_state=RANDOM_SEED)
```

```
[29]: model = tf.keras.models.Sequential([
    tf.keras.layers.Input((21 * 2, )),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(20, activation='relu'),
    tf.keras.layers.Dropout(0.4),
    tf.keras.layers.Dense(10, activation='relu'),
    tf.keras.layers.Dense(NUM_CLASSES, activation='softmax')
])
```

```
[30]: model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dropout_2 (Dropout)	(None, 42)	0
dense_3 (Dense)	(None, 20)	860
dropout_3 (Dropout)	(None, 20)	0
dense_4 (Dense)	(None, 10)	210
dense_5 (Dense)	(None, 26)	286

Total params: 1,356 (5.30 KB)

Trainable params: 1,356 (5.30 KB)

Non-trainable params: 0 (0.00 B)

```
[31]: cp_callback = tf.keras.callbacks.ModelCheckpoint(
      model_save_path, verbose=1, save_weights_only=False)

      es_callback = tf.keras.callbacks.EarlyStopping(patience=20, verbose=1)
```

```
[32]: model.compile(
      optimizer='adam',
      loss='sparse_categorical_crossentropy',
      metrics=['accuracy']
    )
```

```
[33]: history = model.fit(
      X_train,
      y_train,
      epochs=1000,
      batch_size=128,
      validation_data=(X_test, y_test),
      callbacks=[cp_callback, es_callback]
    )
```

Epoch 1/1000

1/5 2s 633ms/step -

accuracy: 0.0078 - loss: 3.3197

Epoch 1: saving model to keypoint_classifier.keras
5/5 1s 55ms/step -
accuracy: 0.0105 - loss: 3.3053 - val_accuracy: 0.0000e+00 - val_loss: 3.2493
Epoch 2/1000
1/5 0s 30ms/step -
accuracy: 0.0156 - loss: 3.2701
Epoch 2: saving model to keypoint_classifier.keras
5/5 0s 24ms/step -
accuracy: 0.0226 - loss: 3.2758 - val_accuracy: 0.0100 - val_loss: 3.2357
Epoch 3/1000
1/5 0s 26ms/step -
accuracy: 0.0234 - loss: 3.2655
Epoch 3: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.0439 - loss: 3.2656 - val_accuracy: 0.0249 - val_loss: 3.2245
Epoch 4/1000
1/5 0s 46ms/step -
accuracy: 0.0469 - loss: 3.2490
Epoch 4: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.0339 - loss: 3.2477 - val_accuracy: 0.0746 - val_loss: 3.2129
Epoch 5/1000
1/5 0s 33ms/step -
accuracy: 0.0312 - loss: 3.2361
Epoch 5: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.0281 - loss: 3.2429 - val_accuracy: 0.0896 - val_loss: 3.2017
Epoch 6/1000
1/5 0s 32ms/step -
accuracy: 0.0703 - loss: 3.2256
Epoch 6: saving model to keypoint_classifier.keras
5/5 0s 24ms/step -
accuracy: 0.0579 - loss: 3.2293 - val_accuracy: 0.1294 - val_loss: 3.1918
Epoch 7/1000
1/5 0s 25ms/step -
accuracy: 0.0781 - loss: 3.2187
Epoch 7: saving model to keypoint_classifier.keras
5/5 0s 24ms/step -
accuracy: 0.0750 - loss: 3.2156 - val_accuracy: 0.1294 - val_loss: 3.1819
Epoch 8/1000
1/5 0s 28ms/step -
accuracy: 0.0625 - loss: 3.2209
Epoch 8: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.0773 - loss: 3.2207 - val_accuracy: 0.1194 - val_loss: 3.1710
Epoch 9/1000
1/5 0s 41ms/step -
accuracy: 0.1016 - loss: 3.1709

Epoch 9: saving model to keypoint_classifier.keras
5/5 0s 34ms/step -
accuracy: 0.0912 - loss: 3.1836 - val_accuracy: 0.1194 - val_loss: 3.1577
Epoch 10/1000
1/5 0s 34ms/step -
accuracy: 0.0859 - loss: 3.1963
Epoch 10: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.0877 - loss: 3.1857 - val_accuracy: 0.1194 - val_loss: 3.1420
Epoch 11/1000
1/5 0s 39ms/step -
accuracy: 0.0859 - loss: 3.1618
Epoch 11: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.0856 - loss: 3.1721 - val_accuracy: 0.1244 - val_loss: 3.1237
Epoch 12/1000
1/5 0s 37ms/step -
accuracy: 0.1016 - loss: 3.1796
Epoch 12: saving model to keypoint_classifier.keras
5/5 0s 35ms/step -
accuracy: 0.1034 - loss: 3.1714 - val_accuracy: 0.1343 - val_loss: 3.1041
Epoch 13/1000
1/5 0s 35ms/step -
accuracy: 0.1172 - loss: 3.1674
Epoch 13: saving model to keypoint_classifier.keras
5/5 0s 34ms/step -
accuracy: 0.1144 - loss: 3.1517 - val_accuracy: 0.1343 - val_loss: 3.0847
Epoch 14/1000
1/5 0s 42ms/step -
accuracy: 0.1094 - loss: 3.1423
Epoch 14: saving model to keypoint_classifier.keras
5/5 0s 31ms/step -
accuracy: 0.1167 - loss: 3.1331 - val_accuracy: 0.1791 - val_loss: 3.0657
Epoch 15/1000
1/5 0s 40ms/step -
accuracy: 0.1328 - loss: 3.1082
Epoch 15: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.1211 - loss: 3.1150 - val_accuracy: 0.1791 - val_loss: 3.0456
Epoch 16/1000
1/5 0s 32ms/step -
accuracy: 0.0938 - loss: 3.1367
Epoch 16: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.1168 - loss: 3.1122 - val_accuracy: 0.1692 - val_loss: 3.0238
Epoch 17/1000
1/5 0s 33ms/step -
accuracy: 0.1094 - loss: 3.0731

Epoch 17: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.1209 - loss: 3.0887 - val_accuracy: 0.1642 - val_loss: 3.0011
Epoch 18/1000
1/5 0s 44ms/step -
accuracy: 0.1016 - loss: 3.0693
Epoch 18: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.1268 - loss: 3.0621 - val_accuracy: 0.1443 - val_loss: 2.9773
Epoch 19/1000
1/5 0s 27ms/step -
accuracy: 0.1094 - loss: 3.0182
Epoch 19: saving model to keypoint_classifier.keras
5/5 0s 35ms/step -
accuracy: 0.1196 - loss: 3.0363 - val_accuracy: 0.1294 - val_loss: 2.9530
Epoch 20/1000
1/5 0s 25ms/step -
accuracy: 0.1094 - loss: 3.0051
Epoch 20: saving model to keypoint_classifier.keras
5/5 0s 36ms/step -
accuracy: 0.1245 - loss: 3.0198 - val_accuracy: 0.1692 - val_loss: 2.9296
Epoch 21/1000
1/5 0s 49ms/step -
accuracy: 0.1719 - loss: 2.9770
Epoch 21: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.1546 - loss: 2.9759 - val_accuracy: 0.1692 - val_loss: 2.9037
Epoch 22/1000
1/5 0s 45ms/step -
accuracy: 0.0781 - loss: 3.0141
Epoch 22: saving model to keypoint_classifier.keras
5/5 0s 31ms/step -
accuracy: 0.1052 - loss: 2.9998 - val_accuracy: 0.1692 - val_loss: 2.8784
Epoch 23/1000
1/5 0s 42ms/step -
accuracy: 0.1953 - loss: 2.8621
Epoch 23: saving model to keypoint_classifier.keras
5/5 0s 36ms/step -
accuracy: 0.1596 - loss: 2.9209 - val_accuracy: 0.1343 - val_loss: 2.8531
Epoch 24/1000
1/5 0s 36ms/step -
accuracy: 0.1172 - loss: 2.9820
Epoch 24: saving model to keypoint_classifier.keras
5/5 0s 34ms/step -
accuracy: 0.1172 - loss: 2.9498 - val_accuracy: 0.1393 - val_loss: 2.8287
Epoch 25/1000
1/5 0s 40ms/step -
accuracy: 0.1250 - loss: 2.9213

Epoch 25: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.1222 - loss: 2.9514 - val_accuracy: 0.1542 - val_loss: 2.8046
Epoch 26/1000
1/5 0s 43ms/step -
accuracy: 0.1328 - loss: 2.9362
Epoch 26: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.1329 - loss: 2.9299 - val_accuracy: 0.2040 - val_loss: 2.7816
Epoch 27/1000
1/5 0s 36ms/step -
accuracy: 0.1797 - loss: 2.8481
Epoch 27: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.1589 - loss: 2.8787 - val_accuracy: 0.2040 - val_loss: 2.7579
Epoch 28/1000
1/5 0s 36ms/step -
accuracy: 0.1250 - loss: 2.8444
Epoch 28: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.1224 - loss: 2.8741 - val_accuracy: 0.1940 - val_loss: 2.7356
Epoch 29/1000
1/5 0s 36ms/step -
accuracy: 0.1016 - loss: 2.8523
Epoch 29: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.1358 - loss: 2.8383 - val_accuracy: 0.1940 - val_loss: 2.7147
Epoch 30/1000
1/5 0s 34ms/step -
accuracy: 0.1250 - loss: 2.7960
Epoch 30: saving model to keypoint_classifier.keras
5/5 0s 34ms/step -
accuracy: 0.1411 - loss: 2.8105 - val_accuracy: 0.1940 - val_loss: 2.6920
Epoch 31/1000
1/5 0s 33ms/step -
accuracy: 0.1172 - loss: 2.8154
Epoch 31: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.1292 - loss: 2.8149 - val_accuracy: 0.1891 - val_loss: 2.6678
Epoch 32/1000
1/5 0s 32ms/step -
accuracy: 0.1484 - loss: 2.7713
Epoch 32: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.1552 - loss: 2.7894 - val_accuracy: 0.1891 - val_loss: 2.6464
Epoch 33/1000
1/5 0s 39ms/step -
accuracy: 0.1328 - loss: 2.8624

Epoch 33: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.1523 - loss: 2.8103 - val_accuracy: 0.1891 - val_loss: 2.6274
Epoch 34/1000
1/5 0s 44ms/step -
accuracy: 0.1719 - loss: 2.6969
Epoch 34: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.1737 - loss: 2.7222 - val_accuracy: 0.1891 - val_loss: 2.6057
Epoch 35/1000
1/5 0s 31ms/step -
accuracy: 0.1797 - loss: 2.7627
Epoch 35: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.1741 - loss: 2.7586 - val_accuracy: 0.1841 - val_loss: 2.5842
Epoch 36/1000
1/5 0s 34ms/step -
accuracy: 0.1250 - loss: 2.7482
Epoch 36: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.1538 - loss: 2.7233 - val_accuracy: 0.1891 - val_loss: 2.5610
Epoch 37/1000
1/5 0s 35ms/step -
accuracy: 0.2109 - loss: 2.6061
Epoch 37: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2053 - loss: 2.6539 - val_accuracy: 0.1891 - val_loss: 2.5387
Epoch 38/1000
1/5 0s 35ms/step -
accuracy: 0.1797 - loss: 2.7123
Epoch 38: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.1915 - loss: 2.6555 - val_accuracy: 0.1990 - val_loss: 2.5164
Epoch 39/1000
1/5 0s 38ms/step -
accuracy: 0.1719 - loss: 2.6449
Epoch 39: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2015 - loss: 2.6445 - val_accuracy: 0.2040 - val_loss: 2.4931
Epoch 40/1000
1/5 0s 35ms/step -
accuracy: 0.1953 - loss: 2.6471
Epoch 40: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.1858 - loss: 2.6170 - val_accuracy: 0.2289 - val_loss: 2.4701
Epoch 41/1000
1/5 0s 46ms/step -
accuracy: 0.1641 - loss: 2.6626

Epoch 41: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.1865 - loss: 2.6096 - val_accuracy: 0.2438 - val_loss: 2.4472
Epoch 42/1000
1/5 0s 40ms/step -
accuracy: 0.2266 - loss: 2.6071
Epoch 42: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2160 - loss: 2.6054 - val_accuracy: 0.2438 - val_loss: 2.4237
Epoch 43/1000
1/5 0s 37ms/step -
accuracy: 0.1641 - loss: 2.6602
Epoch 43: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.1840 - loss: 2.5938 - val_accuracy: 0.2537 - val_loss: 2.4013
Epoch 44/1000
1/5 0s 33ms/step -
accuracy: 0.2891 - loss: 2.4329
Epoch 44: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2256 - loss: 2.5471 - val_accuracy: 0.3134 - val_loss: 2.3802
Epoch 45/1000
1/5 0s 35ms/step -
accuracy: 0.2188 - loss: 2.4855
Epoch 45: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.2060 - loss: 2.5276 - val_accuracy: 0.3284 - val_loss: 2.3610
Epoch 46/1000
1/5 0s 32ms/step -
accuracy: 0.1797 - loss: 2.5098
Epoch 46: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2058 - loss: 2.5289 - val_accuracy: 0.3483 - val_loss: 2.3410
Epoch 47/1000
1/5 0s 34ms/step -
accuracy: 0.2031 - loss: 2.4938
Epoch 47: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2080 - loss: 2.4917 - val_accuracy: 0.3781 - val_loss: 2.3204
Epoch 48/1000
1/5 0s 36ms/step -
accuracy: 0.1953 - loss: 2.5286
Epoch 48: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.1918 - loss: 2.5014 - val_accuracy: 0.3881 - val_loss: 2.2990
Epoch 49/1000
1/5 0s 33ms/step -
accuracy: 0.2578 - loss: 2.4850

Epoch 49: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2473 - loss: 2.4671 - val_accuracy: 0.3930 - val_loss: 2.2779
Epoch 50/1000
1/5 0s 34ms/step -
accuracy: 0.1719 - loss: 2.4548
Epoch 50: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.2055 - loss: 2.4366 - val_accuracy: 0.3881 - val_loss: 2.2588
Epoch 51/1000
1/5 0s 37ms/step -
accuracy: 0.2031 - loss: 2.3839
Epoch 51: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2201 - loss: 2.4048 - val_accuracy: 0.3881 - val_loss: 2.2382
Epoch 52/1000
1/5 0s 35ms/step -
accuracy: 0.2344 - loss: 2.4227
Epoch 52: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2111 - loss: 2.4479 - val_accuracy: 0.4030 - val_loss: 2.2189
Epoch 53/1000
1/5 0s 43ms/step -
accuracy: 0.2266 - loss: 2.4490
Epoch 53: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2277 - loss: 2.4449 - val_accuracy: 0.4179 - val_loss: 2.2030
Epoch 54/1000
1/5 0s 37ms/step -
accuracy: 0.2344 - loss: 2.3852
Epoch 54: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2442 - loss: 2.3725 - val_accuracy: 0.4179 - val_loss: 2.1861
Epoch 55/1000
1/5 0s 34ms/step -
accuracy: 0.2500 - loss: 2.5214
Epoch 55: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2389 - loss: 2.4189 - val_accuracy: 0.4229 - val_loss: 2.1703
Epoch 56/1000
1/5 0s 36ms/step -
accuracy: 0.2500 - loss: 2.3210
Epoch 56: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2439 - loss: 2.3215 - val_accuracy: 0.4129 - val_loss: 2.1545
Epoch 57/1000
1/5 0s 35ms/step -
accuracy: 0.2344 - loss: 2.3858

Epoch 57: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2364 - loss: 2.3682 - val_accuracy: 0.4229 - val_loss: 2.1376
Epoch 58/1000
1/5 0s 36ms/step -
accuracy: 0.2578 - loss: 2.3350
Epoch 58: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2619 - loss: 2.3038 - val_accuracy: 0.4328 - val_loss: 2.1207
Epoch 59/1000
1/5 0s 36ms/step -
accuracy: 0.2891 - loss: 2.2246
Epoch 59: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2556 - loss: 2.3226 - val_accuracy: 0.4279 - val_loss: 2.1048
Epoch 60/1000
1/5 0s 38ms/step -
accuracy: 0.2266 - loss: 2.3056
Epoch 60: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2379 - loss: 2.3140 - val_accuracy: 0.4328 - val_loss: 2.0915
Epoch 61/1000
1/5 0s 41ms/step -
accuracy: 0.2188 - loss: 2.3759
Epoch 61: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2498 - loss: 2.2970 - val_accuracy: 0.4428 - val_loss: 2.0762
Epoch 62/1000
1/5 0s 35ms/step -
accuracy: 0.2344 - loss: 2.2822
Epoch 62: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2512 - loss: 2.2902 - val_accuracy: 0.4527 - val_loss: 2.0625
Epoch 63/1000
1/5 0s 34ms/step -
accuracy: 0.2188 - loss: 2.2934
Epoch 63: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.2532 - loss: 2.2690 - val_accuracy: 0.4726 - val_loss: 2.0479
Epoch 64/1000
1/5 0s 30ms/step -
accuracy: 0.3125 - loss: 2.3185
Epoch 64: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2804 - loss: 2.2552 - val_accuracy: 0.4776 - val_loss: 2.0328
Epoch 65/1000
1/5 0s 38ms/step -
accuracy: 0.2422 - loss: 2.2672

Epoch 65: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2676 - loss: 2.2469 - val_accuracy: 0.4677 - val_loss: 2.0180
Epoch 66/1000
1/5 0s 34ms/step -
accuracy: 0.2969 - loss: 2.2060
Epoch 66: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2612 - loss: 2.2719 - val_accuracy: 0.4627 - val_loss: 2.0050
Epoch 67/1000
1/5 0s 39ms/step -
accuracy: 0.2812 - loss: 2.1692
Epoch 67: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2721 - loss: 2.2139 - val_accuracy: 0.4627 - val_loss: 1.9908
Epoch 68/1000
1/5 0s 35ms/step -
accuracy: 0.2188 - loss: 2.2641
Epoch 68: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2452 - loss: 2.2451 - val_accuracy: 0.4776 - val_loss: 1.9780
Epoch 69/1000
1/5 0s 37ms/step -
accuracy: 0.2500 - loss: 2.2100
Epoch 69: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2668 - loss: 2.1754 - val_accuracy: 0.4677 - val_loss: 1.9648
Epoch 70/1000
1/5 0s 36ms/step -
accuracy: 0.2109 - loss: 2.2779
Epoch 70: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.2536 - loss: 2.2391 - val_accuracy: 0.4776 - val_loss: 1.9536
Epoch 71/1000
1/5 0s 40ms/step -
accuracy: 0.2500 - loss: 2.2414
Epoch 71: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2630 - loss: 2.1823 - val_accuracy: 0.4925 - val_loss: 1.9416
Epoch 72/1000
1/5 0s 32ms/step -
accuracy: 0.2969 - loss: 2.1551
Epoch 72: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2819 - loss: 2.1758 - val_accuracy: 0.5025 - val_loss: 1.9338
Epoch 73/1000
1/5 0s 32ms/step -
accuracy: 0.3672 - loss: 2.0623

Epoch 73: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3189 - loss: 2.1335 - val_accuracy: 0.4975 - val_loss: 1.9270
Epoch 74/1000
1/5 0s 59ms/step -
accuracy: 0.2969 - loss: 2.0770
Epoch 74: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.2780 - loss: 2.1323 - val_accuracy: 0.5075 - val_loss: 1.9167
Epoch 75/1000
1/5 0s 34ms/step -
accuracy: 0.3125 - loss: 2.1228
Epoch 75: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.3029 - loss: 2.1391 - val_accuracy: 0.5075 - val_loss: 1.9072
Epoch 76/1000
1/5 0s 29ms/step -
accuracy: 0.2266 - loss: 2.2008
Epoch 76: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.2622 - loss: 2.1551 - val_accuracy: 0.5075 - val_loss: 1.8956
Epoch 77/1000
1/5 0s 33ms/step -
accuracy: 0.2578 - loss: 2.1162
Epoch 77: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2656 - loss: 2.1163 - val_accuracy: 0.5174 - val_loss: 1.8820
Epoch 78/1000
1/5 0s 33ms/step -
accuracy: 0.3281 - loss: 2.0443
Epoch 78: saving model to keypoint_classifier.keras
5/5 0s 21ms/step -
accuracy: 0.3040 - loss: 2.0747 - val_accuracy: 0.5274 - val_loss: 1.8674
Epoch 79/1000
1/5 0s 36ms/step -
accuracy: 0.1875 - loss: 2.1630
Epoch 79: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.2423 - loss: 2.1503 - val_accuracy: 0.5373 - val_loss: 1.8538
Epoch 80/1000
1/5 0s 34ms/step -
accuracy: 0.2578 - loss: 2.1327
Epoch 80: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2841 - loss: 2.1075 - val_accuracy: 0.5423 - val_loss: 1.8423
Epoch 81/1000
1/5 0s 47ms/step -
accuracy: 0.2812 - loss: 2.1284

Epoch 81: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.2854 - loss: 2.1354 - val_accuracy: 0.5224 - val_loss: 1.8324
Epoch 82/1000
1/5 0s 35ms/step -
accuracy: 0.3047 - loss: 1.9791
Epoch 82: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.2822 - loss: 2.0386 - val_accuracy: 0.5373 - val_loss: 1.8228
Epoch 83/1000
1/5 0s 34ms/step -
accuracy: 0.3047 - loss: 2.0157
Epoch 83: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3129 - loss: 2.0301 - val_accuracy: 0.5373 - val_loss: 1.8146
Epoch 84/1000
1/5 0s 45ms/step -
accuracy: 0.2266 - loss: 2.1062
Epoch 84: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2978 - loss: 2.0806 - val_accuracy: 0.5473 - val_loss: 1.8065
Epoch 85/1000
1/5 0s 37ms/step -
accuracy: 0.2266 - loss: 2.1500
Epoch 85: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2617 - loss: 2.0937 - val_accuracy: 0.5622 - val_loss: 1.7967
Epoch 86/1000
1/5 0s 40ms/step -
accuracy: 0.3047 - loss: 2.0008
Epoch 86: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3072 - loss: 2.0393 - val_accuracy: 0.5522 - val_loss: 1.7871
Epoch 87/1000
1/5 0s 38ms/step -
accuracy: 0.3438 - loss: 1.9286
Epoch 87: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3276 - loss: 2.0026 - val_accuracy: 0.5473 - val_loss: 1.7804
Epoch 88/1000
1/5 0s 41ms/step -
accuracy: 0.3438 - loss: 2.0444
Epoch 88: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3112 - loss: 2.0350 - val_accuracy: 0.5522 - val_loss: 1.7727
Epoch 89/1000
1/5 0s 35ms/step -
accuracy: 0.3594 - loss: 1.9746

Epoch 89: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3312 - loss: 2.0111 - val_accuracy: 0.5572 - val_loss: 1.7672
Epoch 90/1000
1/5 0s 35ms/step -
accuracy: 0.2422 - loss: 2.1615
Epoch 90: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.2950 - loss: 2.0500 - val_accuracy: 0.5373 - val_loss: 1.7607
Epoch 91/1000
1/5 0s 33ms/step -
accuracy: 0.3828 - loss: 1.9848
Epoch 91: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3355 - loss: 2.0233 - val_accuracy: 0.5373 - val_loss: 1.7541
Epoch 92/1000
1/5 0s 35ms/step -
accuracy: 0.3828 - loss: 2.0349
Epoch 92: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.3295 - loss: 2.0350 - val_accuracy: 0.5373 - val_loss: 1.7463
Epoch 93/1000
1/5 0s 32ms/step -
accuracy: 0.3750 - loss: 1.9526
Epoch 93: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3537 - loss: 1.9965 - val_accuracy: 0.5373 - val_loss: 1.7381
Epoch 94/1000
1/5 0s 35ms/step -
accuracy: 0.2891 - loss: 2.1484
Epoch 94: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3202 - loss: 2.0489 - val_accuracy: 0.5423 - val_loss: 1.7317
Epoch 95/1000
1/5 0s 32ms/step -
accuracy: 0.3438 - loss: 1.9764
Epoch 95: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3302 - loss: 1.9772 - val_accuracy: 0.5473 - val_loss: 1.7224
Epoch 96/1000
1/5 0s 31ms/step -
accuracy: 0.3359 - loss: 1.9897
Epoch 96: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3413 - loss: 1.9724 - val_accuracy: 0.5473 - val_loss: 1.7108
Epoch 97/1000
1/5 0s 35ms/step -
accuracy: 0.2500 - loss: 2.0835

Epoch 97: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3078 - loss: 2.0195 - val_accuracy: 0.5473 - val_loss: 1.7030
Epoch 98/1000
1/5 0s 42ms/step -
accuracy: 0.3203 - loss: 1.9652
Epoch 98: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3186 - loss: 2.0114 - val_accuracy: 0.5473 - val_loss: 1.6950
Epoch 99/1000
1/5 0s 33ms/step -
accuracy: 0.3359 - loss: 1.9997
Epoch 99: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3550 - loss: 1.9489 - val_accuracy: 0.5473 - val_loss: 1.6868
Epoch 100/1000
1/5 0s 38ms/step -
accuracy: 0.2812 - loss: 2.0548
Epoch 100: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.2988 - loss: 2.0086 - val_accuracy: 0.5473 - val_loss: 1.6776
Epoch 101/1000
1/5 0s 35ms/step -
accuracy: 0.3516 - loss: 1.9923
Epoch 101: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3378 - loss: 1.9782 - val_accuracy: 0.5473 - val_loss: 1.6692
Epoch 102/1000
1/5 0s 33ms/step -
accuracy: 0.3594 - loss: 1.9691
Epoch 102: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3425 - loss: 1.9593 - val_accuracy: 0.5473 - val_loss: 1.6599
Epoch 103/1000
1/5 0s 34ms/step -
accuracy: 0.3828 - loss: 1.9438
Epoch 103: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3522 - loss: 1.9447 - val_accuracy: 0.5473 - val_loss: 1.6488
Epoch 104/1000
1/5 0s 38ms/step -
accuracy: 0.3203 - loss: 2.0122
Epoch 104: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3249 - loss: 1.9542 - val_accuracy: 0.5473 - val_loss: 1.6388
Epoch 105/1000
1/5 0s 43ms/step -
accuracy: 0.3594 - loss: 2.0289

Epoch 105: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.3371 - loss: 1.9736 - val_accuracy: 0.5572 - val_loss: 1.6331
Epoch 106/1000
1/5 0s 34ms/step -
accuracy: 0.4062 - loss: 1.9104
Epoch 106: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3496 - loss: 1.9600 - val_accuracy: 0.5572 - val_loss: 1.6271
Epoch 107/1000
1/5 0s 34ms/step -
accuracy: 0.3828 - loss: 1.8570
Epoch 107: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3515 - loss: 1.8858 - val_accuracy: 0.5522 - val_loss: 1.6208
Epoch 108/1000
1/5 0s 37ms/step -
accuracy: 0.4141 - loss: 1.8922
Epoch 108: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3889 - loss: 1.8828 - val_accuracy: 0.5473 - val_loss: 1.6134
Epoch 109/1000
1/5 0s 31ms/step -
accuracy: 0.3359 - loss: 1.9626
Epoch 109: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3409 - loss: 1.9184 - val_accuracy: 0.5473 - val_loss: 1.6067
Epoch 110/1000
1/5 0s 33ms/step -
accuracy: 0.3125 - loss: 1.9304
Epoch 110: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3269 - loss: 1.9227 - val_accuracy: 0.5572 - val_loss: 1.6023
Epoch 111/1000
1/5 0s 34ms/step -
accuracy: 0.3125 - loss: 1.9344
Epoch 111: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3422 - loss: 1.9094 - val_accuracy: 0.5622 - val_loss: 1.5985
Epoch 112/1000
1/5 0s 34ms/step -
accuracy: 0.3359 - loss: 1.9213
Epoch 112: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3457 - loss: 1.9176 - val_accuracy: 0.5771 - val_loss: 1.5957
Epoch 113/1000
1/5 0s 38ms/step -
accuracy: 0.2734 - loss: 1.9302

Epoch 113: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3155 - loss: 1.9079 - val_accuracy: 0.5721 - val_loss: 1.5925
Epoch 114/1000
1/5 0s 31ms/step -
accuracy: 0.3203 - loss: 1.8906
Epoch 114: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3533 - loss: 1.8654 - val_accuracy: 0.5721 - val_loss: 1.5850
Epoch 115/1000
1/5 0s 37ms/step -
accuracy: 0.3984 - loss: 1.8721
Epoch 115: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.3677 - loss: 1.9000 - val_accuracy: 0.5672 - val_loss: 1.5767
Epoch 116/1000
1/5 0s 39ms/step -
accuracy: 0.3047 - loss: 1.9780
Epoch 116: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3463 - loss: 1.9072 - val_accuracy: 0.5672 - val_loss: 1.5695
Epoch 117/1000
1/5 0s 35ms/step -
accuracy: 0.3984 - loss: 1.9115
Epoch 117: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3773 - loss: 1.8800 - val_accuracy: 0.5771 - val_loss: 1.5616
Epoch 118/1000
1/5 0s 34ms/step -
accuracy: 0.3984 - loss: 1.8418
Epoch 118: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3641 - loss: 1.8522 - val_accuracy: 0.5672 - val_loss: 1.5514
Epoch 119/1000
1/5 0s 36ms/step -
accuracy: 0.3203 - loss: 1.9552
Epoch 119: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3290 - loss: 1.9208 - val_accuracy: 0.5672 - val_loss: 1.5443
Epoch 120/1000
1/5 0s 32ms/step -
accuracy: 0.2969 - loss: 2.0292
Epoch 120: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3379 - loss: 1.9287 - val_accuracy: 0.5622 - val_loss: 1.5404
Epoch 121/1000
1/5 0s 57ms/step -
accuracy: 0.3281 - loss: 1.9189

Epoch 121: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3636 - loss: 1.8682 - val_accuracy: 0.5672 - val_loss: 1.5375
Epoch 122/1000
1/5 0s 34ms/step -
accuracy: 0.3203 - loss: 1.8959
Epoch 122: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.3486 - loss: 1.8571 - val_accuracy: 0.5821 - val_loss: 1.5310
Epoch 123/1000
1/5 0s 34ms/step -
accuracy: 0.4219 - loss: 1.7765
Epoch 123: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4024 - loss: 1.8078 - val_accuracy: 0.5970 - val_loss: 1.5218
Epoch 124/1000
1/5 0s 35ms/step -
accuracy: 0.3828 - loss: 1.8599
Epoch 124: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3614 - loss: 1.8842 - val_accuracy: 0.5821 - val_loss: 1.5130
Epoch 125/1000
1/5 0s 32ms/step -
accuracy: 0.3906 - loss: 1.9327
Epoch 125: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3802 - loss: 1.8785 - val_accuracy: 0.5821 - val_loss: 1.5057
Epoch 126/1000
1/5 0s 34ms/step -
accuracy: 0.4141 - loss: 1.8076
Epoch 126: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3811 - loss: 1.8507 - val_accuracy: 0.5821 - val_loss: 1.5004
Epoch 127/1000
1/5 0s 33ms/step -
accuracy: 0.3984 - loss: 1.8120
Epoch 127: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3705 - loss: 1.8466 - val_accuracy: 0.5771 - val_loss: 1.4948
Epoch 128/1000
1/5 0s 34ms/step -
accuracy: 0.3906 - loss: 1.8493
Epoch 128: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3865 - loss: 1.8468 - val_accuracy: 0.5771 - val_loss: 1.4915
Epoch 129/1000
1/5 0s 33ms/step -
accuracy: 0.3594 - loss: 1.7967

Epoch 129: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3576 - loss: 1.8072 - val_accuracy: 0.5771 - val_loss: 1.4850
Epoch 130/1000
1/5 0s 35ms/step -
accuracy: 0.4141 - loss: 1.7797
Epoch 130: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3677 - loss: 1.8405 - val_accuracy: 0.5871 - val_loss: 1.4800
Epoch 131/1000
1/5 0s 30ms/step -
accuracy: 0.3047 - loss: 1.9965
Epoch 131: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3413 - loss: 1.8698 - val_accuracy: 0.5771 - val_loss: 1.4754
Epoch 132/1000
1/5 0s 36ms/step -
accuracy: 0.4688 - loss: 1.6428
Epoch 132: saving model to keypoint_classifier.keras
5/5 0s 41ms/step -
accuracy: 0.4047 - loss: 1.7695 - val_accuracy: 0.5771 - val_loss: 1.4705
Epoch 133/1000
1/5 0s 39ms/step -
accuracy: 0.3906 - loss: 1.8152
Epoch 133: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3671 - loss: 1.8405 - val_accuracy: 0.5821 - val_loss: 1.4676
Epoch 134/1000
1/5 0s 36ms/step -
accuracy: 0.3281 - loss: 1.8439
Epoch 134: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3562 - loss: 1.8186 - val_accuracy: 0.5721 - val_loss: 1.4659
Epoch 135/1000
1/5 0s 32ms/step -
accuracy: 0.3828 - loss: 1.8175
Epoch 135: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3886 - loss: 1.7690 - val_accuracy: 0.5821 - val_loss: 1.4637
Epoch 136/1000
1/5 0s 31ms/step -
accuracy: 0.3672 - loss: 1.7196
Epoch 136: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3720 - loss: 1.7918 - val_accuracy: 0.5821 - val_loss: 1.4573
Epoch 137/1000
1/5 0s 30ms/step -
accuracy: 0.3984 - loss: 1.8338

Epoch 137: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3892 - loss: 1.8268 - val_accuracy: 0.5821 - val_loss: 1.4511
Epoch 138/1000
1/5 0s 35ms/step -
accuracy: 0.4453 - loss: 1.6614
Epoch 138: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4080 - loss: 1.7380 - val_accuracy: 0.5821 - val_loss: 1.4444
Epoch 139/1000
1/5 0s 36ms/step -
accuracy: 0.3516 - loss: 1.8403
Epoch 139: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3490 - loss: 1.8305 - val_accuracy: 0.5920 - val_loss: 1.4395
Epoch 140/1000
1/5 0s 33ms/step -
accuracy: 0.3750 - loss: 1.8765
Epoch 140: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4009 - loss: 1.8080 - val_accuracy: 0.6020 - val_loss: 1.4359
Epoch 141/1000
1/5 0s 33ms/step -
accuracy: 0.4297 - loss: 1.7004
Epoch 141: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4119 - loss: 1.7158 - val_accuracy: 0.5970 - val_loss: 1.4313
Epoch 142/1000
1/5 0s 37ms/step -
accuracy: 0.4062 - loss: 1.7733
Epoch 142: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4079 - loss: 1.7541 - val_accuracy: 0.5970 - val_loss: 1.4228
Epoch 143/1000
1/5 0s 34ms/step -
accuracy: 0.4375 - loss: 1.7276
Epoch 143: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4091 - loss: 1.7861 - val_accuracy: 0.5920 - val_loss: 1.4172
Epoch 144/1000
1/5 0s 38ms/step -
accuracy: 0.2891 - loss: 1.8453
Epoch 144: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3246 - loss: 1.8417 - val_accuracy: 0.6020 - val_loss: 1.4112
Epoch 145/1000
1/5 0s 34ms/step -
accuracy: 0.3984 - loss: 1.7780

Epoch 145: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3697 - loss: 1.7845 - val_accuracy: 0.6070 - val_loss: 1.4071
Epoch 146/1000
1/5 0s 33ms/step -
accuracy: 0.3750 - loss: 1.7996
Epoch 146: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3927 - loss: 1.7741 - val_accuracy: 0.6119 - val_loss: 1.4001
Epoch 147/1000
1/5 0s 33ms/step -
accuracy: 0.4531 - loss: 1.7022
Epoch 147: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4239 - loss: 1.7486 - val_accuracy: 0.6119 - val_loss: 1.3914
Epoch 148/1000
1/5 0s 39ms/step -
accuracy: 0.3594 - loss: 1.7188
Epoch 148: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.3624 - loss: 1.7248 - val_accuracy: 0.6020 - val_loss: 1.3830
Epoch 149/1000
1/5 0s 34ms/step -
accuracy: 0.4297 - loss: 1.6219
Epoch 149: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3960 - loss: 1.7282 - val_accuracy: 0.5920 - val_loss: 1.3758
Epoch 150/1000
1/5 0s 35ms/step -
accuracy: 0.3438 - loss: 1.8197
Epoch 150: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3929 - loss: 1.7558 - val_accuracy: 0.5771 - val_loss: 1.3709
Epoch 151/1000
1/5 0s 36ms/step -
accuracy: 0.3750 - loss: 1.8748
Epoch 151: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3797 - loss: 1.8087 - val_accuracy: 0.5721 - val_loss: 1.3662
Epoch 152/1000
1/5 0s 37ms/step -
accuracy: 0.3672 - loss: 1.7996
Epoch 152: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3749 - loss: 1.7529 - val_accuracy: 0.5721 - val_loss: 1.3612
Epoch 153/1000
1/5 0s 34ms/step -
accuracy: 0.4062 - loss: 1.7030

Epoch 153: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3924 - loss: 1.7511 - val_accuracy: 0.5771 - val_loss: 1.3566
Epoch 154/1000
1/5 0s 39ms/step -
accuracy: 0.4219 - loss: 1.7681
Epoch 154: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3901 - loss: 1.7627 - val_accuracy: 0.5920 - val_loss: 1.3522
Epoch 155/1000
1/5 0s 36ms/step -
accuracy: 0.3906 - loss: 1.7420
Epoch 155: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3955 - loss: 1.7276 - val_accuracy: 0.6070 - val_loss: 1.3491
Epoch 156/1000
1/5 0s 30ms/step -
accuracy: 0.3672 - loss: 1.8017
Epoch 156: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.3714 - loss: 1.7551 - val_accuracy: 0.6119 - val_loss: 1.3467
Epoch 157/1000
1/5 0s 33ms/step -
accuracy: 0.3984 - loss: 1.8117
Epoch 157: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4249 - loss: 1.7327 - val_accuracy: 0.6219 - val_loss: 1.3443
Epoch 158/1000
1/5 0s 32ms/step -
accuracy: 0.4453 - loss: 1.7173
Epoch 158: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4229 - loss: 1.7240 - val_accuracy: 0.6169 - val_loss: 1.3370
Epoch 159/1000
1/5 0s 35ms/step -
accuracy: 0.4297 - loss: 1.7228
Epoch 159: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4167 - loss: 1.7355 - val_accuracy: 0.6020 - val_loss: 1.3330
Epoch 160/1000
1/5 0s 33ms/step -
accuracy: 0.3672 - loss: 1.7321
Epoch 160: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3874 - loss: 1.7188 - val_accuracy: 0.6020 - val_loss: 1.3250
Epoch 161/1000
1/5 0s 39ms/step -
accuracy: 0.5156 - loss: 1.6060

Epoch 161: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4388 - loss: 1.6673 - val_accuracy: 0.6070 - val_loss: 1.3159
Epoch 162/1000
1/5 0s 34ms/step -
accuracy: 0.4766 - loss: 1.6640
Epoch 162: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4291 - loss: 1.6923 - val_accuracy: 0.6070 - val_loss: 1.3082
Epoch 163/1000
1/5 0s 34ms/step -
accuracy: 0.3281 - loss: 1.7801
Epoch 163: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.3841 - loss: 1.7370 - val_accuracy: 0.5970 - val_loss: 1.3026
Epoch 164/1000
1/5 0s 39ms/step -
accuracy: 0.3906 - loss: 1.7378
Epoch 164: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.3873 - loss: 1.7344 - val_accuracy: 0.6020 - val_loss: 1.2996
Epoch 165/1000
1/5 0s 32ms/step -
accuracy: 0.3828 - loss: 1.8086
Epoch 165: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4081 - loss: 1.7271 - val_accuracy: 0.6020 - val_loss: 1.2955
Epoch 166/1000
1/5 0s 38ms/step -
accuracy: 0.4062 - loss: 1.6254
Epoch 166: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4162 - loss: 1.6736 - val_accuracy: 0.6219 - val_loss: 1.2894
Epoch 167/1000
1/5 0s 34ms/step -
accuracy: 0.3828 - loss: 1.7883
Epoch 167: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4034 - loss: 1.7116 - val_accuracy: 0.6219 - val_loss: 1.2834
Epoch 168/1000
1/5 0s 40ms/step -
accuracy: 0.4688 - loss: 1.5814
Epoch 168: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.4176 - loss: 1.6819 - val_accuracy: 0.6269 - val_loss: 1.2836
Epoch 169/1000
1/5 0s 26ms/step -
accuracy: 0.5000 - loss: 1.6242

Epoch 169: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4569 - loss: 1.6408 - val_accuracy: 0.6269 - val_loss: 1.2823
Epoch 170/1000
1/5 0s 33ms/step -
accuracy: 0.4062 - loss: 1.6861
Epoch 170: saving model to keypoint_classifier.keras
5/5 0s 35ms/step -
accuracy: 0.4206 - loss: 1.6789 - val_accuracy: 0.6169 - val_loss: 1.2800
Epoch 171/1000
1/5 0s 27ms/step -
accuracy: 0.4531 - loss: 1.6658
Epoch 171: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4336 - loss: 1.6871 - val_accuracy: 0.6070 - val_loss: 1.2744
Epoch 172/1000
1/5 0s 35ms/step -
accuracy: 0.3672 - loss: 1.8278
Epoch 172: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.3900 - loss: 1.7389 - val_accuracy: 0.5970 - val_loss: 1.2643
Epoch 173/1000
1/5 0s 35ms/step -
accuracy: 0.4531 - loss: 1.5453
Epoch 173: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4224 - loss: 1.6510 - val_accuracy: 0.5920 - val_loss: 1.2586
Epoch 174/1000
1/5 0s 34ms/step -
accuracy: 0.4609 - loss: 1.6403
Epoch 174: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4391 - loss: 1.6494 - val_accuracy: 0.6020 - val_loss: 1.2557
Epoch 175/1000
1/5 0s 36ms/step -
accuracy: 0.3516 - loss: 1.7207
Epoch 175: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.3838 - loss: 1.6980 - val_accuracy: 0.6219 - val_loss: 1.2524
Epoch 176/1000
1/5 0s 33ms/step -
accuracy: 0.4844 - loss: 1.6320
Epoch 176: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4495 - loss: 1.6551 - val_accuracy: 0.6269 - val_loss: 1.2474
Epoch 177/1000
1/5 0s 36ms/step -
accuracy: 0.4141 - loss: 1.5362

Epoch 177: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4166 - loss: 1.6330 - val_accuracy: 0.6318 - val_loss: 1.2446
Epoch 178/1000
1/5 0s 33ms/step -
accuracy: 0.3516 - loss: 1.7478
Epoch 178: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4001 - loss: 1.6776 - val_accuracy: 0.6269 - val_loss: 1.2451
Epoch 179/1000
1/5 0s 34ms/step -
accuracy: 0.4688 - loss: 1.5905
Epoch 179: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4580 - loss: 1.6123 - val_accuracy: 0.6219 - val_loss: 1.2418
Epoch 180/1000
1/5 0s 38ms/step -
accuracy: 0.4531 - loss: 1.6462
Epoch 180: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4256 - loss: 1.6552 - val_accuracy: 0.5970 - val_loss: 1.2362
Epoch 181/1000
1/5 0s 33ms/step -
accuracy: 0.4766 - loss: 1.5482
Epoch 181: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4613 - loss: 1.5921 - val_accuracy: 0.6169 - val_loss: 1.2262
Epoch 182/1000
1/5 0s 33ms/step -
accuracy: 0.4141 - loss: 1.7045
Epoch 182: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4380 - loss: 1.6462 - val_accuracy: 0.6219 - val_loss: 1.2171
Epoch 183/1000
1/5 0s 33ms/step -
accuracy: 0.3828 - loss: 1.7433
Epoch 183: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4086 - loss: 1.6820 - val_accuracy: 0.6368 - val_loss: 1.2098
Epoch 184/1000
1/5 0s 31ms/step -
accuracy: 0.4609 - loss: 1.6665
Epoch 184: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4694 - loss: 1.6237 - val_accuracy: 0.6418 - val_loss: 1.2044
Epoch 185/1000
1/5 0s 35ms/step -
accuracy: 0.4922 - loss: 1.5895

Epoch 185: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4685 - loss: 1.5825 - val_accuracy: 0.6468 - val_loss: 1.2019
Epoch 186/1000
1/5 0s 33ms/step -
accuracy: 0.5469 - loss: 1.5107
Epoch 186: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4855 - loss: 1.5668 - val_accuracy: 0.6567 - val_loss: 1.1996
Epoch 187/1000
1/5 0s 29ms/step -
accuracy: 0.3750 - loss: 1.5787
Epoch 187: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4186 - loss: 1.6237 - val_accuracy: 0.6517 - val_loss: 1.2024
Epoch 188/1000
1/5 0s 34ms/step -
accuracy: 0.4922 - loss: 1.5535
Epoch 188: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4622 - loss: 1.6103 - val_accuracy: 0.6418 - val_loss: 1.2066
Epoch 189/1000
1/5 0s 37ms/step -
accuracy: 0.3828 - loss: 1.8023
Epoch 189: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4154 - loss: 1.6959 - val_accuracy: 0.6368 - val_loss: 1.2069
Epoch 190/1000
1/5 0s 26ms/step -
accuracy: 0.4219 - loss: 1.6481
Epoch 190: saving model to keypoint_classifier.keras
5/5 0s 31ms/step -
accuracy: 0.4252 - loss: 1.6371 - val_accuracy: 0.6318 - val_loss: 1.2047
Epoch 191/1000
1/5 0s 33ms/step -
accuracy: 0.4453 - loss: 1.5548
Epoch 191: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4139 - loss: 1.6217 - val_accuracy: 0.6368 - val_loss: 1.2026
Epoch 192/1000
1/5 0s 35ms/step -
accuracy: 0.4141 - loss: 1.5918
Epoch 192: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4434 - loss: 1.5886 - val_accuracy: 0.6318 - val_loss: 1.1982
Epoch 193/1000
1/5 0s 41ms/step -
accuracy: 0.4453 - loss: 1.7044

Epoch 193: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4454 - loss: 1.6414 - val_accuracy: 0.6318 - val_loss: 1.1979
Epoch 194/1000
1/5 0s 33ms/step -
accuracy: 0.4922 - loss: 1.5592
Epoch 194: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4608 - loss: 1.5926 - val_accuracy: 0.6169 - val_loss: 1.1952
Epoch 195/1000
1/5 0s 35ms/step -
accuracy: 0.5234 - loss: 1.6353
Epoch 195: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4771 - loss: 1.6059 - val_accuracy: 0.6318 - val_loss: 1.1886
Epoch 196/1000
1/5 0s 32ms/step -
accuracy: 0.4922 - loss: 1.4985
Epoch 196: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4530 - loss: 1.5619 - val_accuracy: 0.6418 - val_loss: 1.1807
Epoch 197/1000
1/5 0s 33ms/step -
accuracy: 0.4688 - loss: 1.5534
Epoch 197: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4460 - loss: 1.5694 - val_accuracy: 0.6418 - val_loss: 1.1740
Epoch 198/1000
1/5 0s 48ms/step -
accuracy: 0.3984 - loss: 1.6654
Epoch 198: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4282 - loss: 1.6061 - val_accuracy: 0.6667 - val_loss: 1.1674
Epoch 199/1000
1/5 0s 42ms/step -
accuracy: 0.4844 - loss: 1.5338
Epoch 199: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.4462 - loss: 1.6086 - val_accuracy: 0.6667 - val_loss: 1.1625
Epoch 200/1000
1/5 0s 36ms/step -
accuracy: 0.5156 - loss: 1.5053
Epoch 200: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4717 - loss: 1.5600 - val_accuracy: 0.6617 - val_loss: 1.1617
Epoch 201/1000
1/5 0s 36ms/step -
accuracy: 0.4844 - loss: 1.5406

Epoch 201: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4911 - loss: 1.5351 - val_accuracy: 0.6667 - val_loss: 1.1619
Epoch 202/1000
1/5 0s 35ms/step -
accuracy: 0.4375 - loss: 1.6248
Epoch 202: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4344 - loss: 1.6454 - val_accuracy: 0.6617 - val_loss: 1.1598
Epoch 203/1000
1/5 0s 42ms/step -
accuracy: 0.4453 - loss: 1.5789
Epoch 203: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4417 - loss: 1.5983 - val_accuracy: 0.6567 - val_loss: 1.1588
Epoch 204/1000
1/5 0s 31ms/step -
accuracy: 0.3516 - loss: 1.6761
Epoch 204: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4016 - loss: 1.6335 - val_accuracy: 0.6368 - val_loss: 1.1573
Epoch 205/1000
1/5 0s 37ms/step -
accuracy: 0.4062 - loss: 1.6365
Epoch 205: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4365 - loss: 1.6071 - val_accuracy: 0.6368 - val_loss: 1.1564
Epoch 206/1000
1/5 0s 43ms/step -
accuracy: 0.4453 - loss: 1.5934
Epoch 206: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4473 - loss: 1.5990 - val_accuracy: 0.6219 - val_loss: 1.1546
Epoch 207/1000
1/5 0s 34ms/step -
accuracy: 0.4609 - loss: 1.5736
Epoch 207: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4655 - loss: 1.5820 - val_accuracy: 0.6269 - val_loss: 1.1510
Epoch 208/1000
1/5 0s 38ms/step -
accuracy: 0.3906 - loss: 1.6762
Epoch 208: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4323 - loss: 1.6066 - val_accuracy: 0.6318 - val_loss: 1.1462
Epoch 209/1000
1/5 0s 38ms/step -
accuracy: 0.4531 - loss: 1.6215

Epoch 209: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4439 - loss: 1.6037 - val_accuracy: 0.6418 - val_loss: 1.1448
Epoch 210/1000
1/5 0s 33ms/step -
accuracy: 0.4062 - loss: 1.5679
Epoch 210: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4218 - loss: 1.6069 - val_accuracy: 0.6418 - val_loss: 1.1459
Epoch 211/1000
1/5 0s 36ms/step -
accuracy: 0.4531 - loss: 1.5614
Epoch 211: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4264 - loss: 1.5997 - val_accuracy: 0.6517 - val_loss: 1.1445
Epoch 212/1000
1/5 0s 37ms/step -
accuracy: 0.4688 - loss: 1.4559
Epoch 212: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4556 - loss: 1.5030 - val_accuracy: 0.6766 - val_loss: 1.1418
Epoch 213/1000
1/5 0s 40ms/step -
accuracy: 0.4219 - loss: 1.6374
Epoch 213: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4719 - loss: 1.5691 - val_accuracy: 0.6667 - val_loss: 1.1368
Epoch 214/1000
1/5 0s 33ms/step -
accuracy: 0.5938 - loss: 1.4591
Epoch 214: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5246 - loss: 1.4893 - val_accuracy: 0.6567 - val_loss: 1.1317
Epoch 215/1000
1/5 0s 35ms/step -
accuracy: 0.3984 - loss: 1.6227
Epoch 215: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4218 - loss: 1.6020 - val_accuracy: 0.6468 - val_loss: 1.1279
Epoch 216/1000
1/5 0s 34ms/step -
accuracy: 0.4609 - loss: 1.5930
Epoch 216: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.4750 - loss: 1.5528 - val_accuracy: 0.6517 - val_loss: 1.1229
Epoch 217/1000
1/5 0s 35ms/step -
accuracy: 0.4453 - loss: 1.5493

Epoch 217: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4443 - loss: 1.5988 - val_accuracy: 0.6617 - val_loss: 1.1189
Epoch 218/1000
1/5 0s 53ms/step -
accuracy: 0.4453 - loss: 1.4713
Epoch 218: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.4746 - loss: 1.4951 - val_accuracy: 0.6617 - val_loss: 1.1185
Epoch 219/1000
1/5 0s 36ms/step -
accuracy: 0.4297 - loss: 1.6431
Epoch 219: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4464 - loss: 1.5926 - val_accuracy: 0.6667 - val_loss: 1.1175
Epoch 220/1000
1/5 0s 35ms/step -
accuracy: 0.4375 - loss: 1.6605
Epoch 220: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4534 - loss: 1.6111 - val_accuracy: 0.6667 - val_loss: 1.1148
Epoch 221/1000
1/5 0s 46ms/step -
accuracy: 0.4766 - loss: 1.5506
Epoch 221: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4894 - loss: 1.5401 - val_accuracy: 0.6667 - val_loss: 1.1124
Epoch 222/1000
1/5 0s 41ms/step -
accuracy: 0.4453 - loss: 1.5686
Epoch 222: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.4503 - loss: 1.5457 - val_accuracy: 0.6716 - val_loss: 1.1117
Epoch 223/1000
1/5 0s 40ms/step -
accuracy: 0.4766 - loss: 1.5223
Epoch 223: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4640 - loss: 1.5133 - val_accuracy: 0.6667 - val_loss: 1.1098
Epoch 224/1000
1/5 0s 45ms/step -
accuracy: 0.4219 - loss: 1.5648
Epoch 224: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4679 - loss: 1.5204 - val_accuracy: 0.6716 - val_loss: 1.1073
Epoch 225/1000
1/5 0s 47ms/step -
accuracy: 0.4375 - loss: 1.5438

Epoch 225: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4496 - loss: 1.5180 - val_accuracy: 0.6667 - val_loss: 1.1025
Epoch 226/1000
1/5 0s 35ms/step -
accuracy: 0.4688 - loss: 1.5018
Epoch 226: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.4755 - loss: 1.5142 - val_accuracy: 0.6766 - val_loss: 1.0977
Epoch 227/1000
1/5 0s 37ms/step -
accuracy: 0.4141 - loss: 1.6859
Epoch 227: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4527 - loss: 1.5947 - val_accuracy: 0.6716 - val_loss: 1.0936
Epoch 228/1000
1/5 0s 43ms/step -
accuracy: 0.4219 - loss: 1.6402
Epoch 228: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4538 - loss: 1.5599 - val_accuracy: 0.6517 - val_loss: 1.0925
Epoch 229/1000
1/5 0s 35ms/step -
accuracy: 0.4531 - loss: 1.5938
Epoch 229: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4563 - loss: 1.5851 - val_accuracy: 0.6468 - val_loss: 1.0930
Epoch 230/1000
1/5 0s 36ms/step -
accuracy: 0.4844 - loss: 1.4871
Epoch 230: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4607 - loss: 1.5268 - val_accuracy: 0.6517 - val_loss: 1.0941
Epoch 231/1000
1/5 0s 38ms/step -
accuracy: 0.3281 - loss: 1.8414
Epoch 231: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4147 - loss: 1.6522 - val_accuracy: 0.6567 - val_loss: 1.0941
Epoch 232/1000
1/5 0s 33ms/step -
accuracy: 0.5000 - loss: 1.5200
Epoch 232: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4944 - loss: 1.4924 - val_accuracy: 0.6667 - val_loss: 1.0917
Epoch 233/1000
1/5 0s 46ms/step -
accuracy: 0.5234 - loss: 1.4128

Epoch 233: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4850 - loss: 1.5005 - val_accuracy: 0.6667 - val_loss: 1.0905
Epoch 234/1000
1/5 0s 44ms/step -
accuracy: 0.4375 - loss: 1.5827
Epoch 234: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4644 - loss: 1.5351 - val_accuracy: 0.6667 - val_loss: 1.0863
Epoch 235/1000
1/5 0s 32ms/step -
accuracy: 0.5469 - loss: 1.5449
Epoch 235: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5136 - loss: 1.5356 - val_accuracy: 0.6667 - val_loss: 1.0795
Epoch 236/1000
1/5 0s 34ms/step -
accuracy: 0.4531 - loss: 1.5115
Epoch 236: saving model to keypoint_classifier.keras
5/5 0s 35ms/step -
accuracy: 0.4532 - loss: 1.5212 - val_accuracy: 0.6617 - val_loss: 1.0735
Epoch 237/1000
1/5 0s 40ms/step -
accuracy: 0.4297 - loss: 1.5654
Epoch 237: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4527 - loss: 1.5304 - val_accuracy: 0.6567 - val_loss: 1.0687
Epoch 238/1000
1/5 0s 34ms/step -
accuracy: 0.4766 - loss: 1.4860
Epoch 238: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4748 - loss: 1.5191 - val_accuracy: 0.6617 - val_loss: 1.0669
Epoch 239/1000
1/5 0s 37ms/step -
accuracy: 0.4688 - loss: 1.3988
Epoch 239: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4597 - loss: 1.4910 - val_accuracy: 0.6617 - val_loss: 1.0638
Epoch 240/1000
1/5 0s 32ms/step -
accuracy: 0.5000 - loss: 1.4746
Epoch 240: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4678 - loss: 1.5269 - val_accuracy: 0.6617 - val_loss: 1.0627
Epoch 241/1000
1/5 0s 32ms/step -
accuracy: 0.4219 - loss: 1.4681

Epoch 241: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4673 - loss: 1.4949 - val_accuracy: 0.6816 - val_loss: 1.0638
Epoch 242/1000
1/5 0s 36ms/step -
accuracy: 0.4531 - loss: 1.4668
Epoch 242: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4721 - loss: 1.4680 - val_accuracy: 0.6766 - val_loss: 1.0640
Epoch 243/1000
1/5 0s 32ms/step -
accuracy: 0.4844 - loss: 1.5729
Epoch 243: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4735 - loss: 1.5737 - val_accuracy: 0.6567 - val_loss: 1.0629
Epoch 244/1000
1/5 0s 37ms/step -
accuracy: 0.4766 - loss: 1.3490
Epoch 244: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4841 - loss: 1.4155 - val_accuracy: 0.6517 - val_loss: 1.0613
Epoch 245/1000
1/5 0s 37ms/step -
accuracy: 0.4531 - loss: 1.6104
Epoch 245: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4401 - loss: 1.5778 - val_accuracy: 0.6617 - val_loss: 1.0603
Epoch 246/1000
1/5 0s 31ms/step -
accuracy: 0.5234 - loss: 1.3781
Epoch 246: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4856 - loss: 1.4263 - val_accuracy: 0.6617 - val_loss: 1.0581
Epoch 247/1000
1/5 0s 38ms/step -
accuracy: 0.4531 - loss: 1.4567
Epoch 247: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4745 - loss: 1.4814 - val_accuracy: 0.6667 - val_loss: 1.0525
Epoch 248/1000
1/5 0s 34ms/step -
accuracy: 0.4297 - loss: 1.6021
Epoch 248: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4675 - loss: 1.5417 - val_accuracy: 0.6617 - val_loss: 1.0513
Epoch 249/1000
1/5 0s 32ms/step -
accuracy: 0.5000 - loss: 1.4521

Epoch 249: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4680 - loss: 1.5013 - val_accuracy: 0.6567 - val_loss: 1.0471
Epoch 250/1000
1/5 0s 44ms/step -
accuracy: 0.5391 - loss: 1.4818
Epoch 250: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5127 - loss: 1.4841 - val_accuracy: 0.6517 - val_loss: 1.0420
Epoch 251/1000
1/5 0s 33ms/step -
accuracy: 0.4609 - loss: 1.3355
Epoch 251: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4837 - loss: 1.4145 - val_accuracy: 0.6766 - val_loss: 1.0352
Epoch 252/1000
1/5 0s 41ms/step -
accuracy: 0.5078 - loss: 1.3741
Epoch 252: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4817 - loss: 1.4417 - val_accuracy: 0.6766 - val_loss: 1.0342
Epoch 253/1000
1/5 0s 45ms/step -
accuracy: 0.4375 - loss: 1.5925
Epoch 253: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.4499 - loss: 1.5701 - val_accuracy: 0.6716 - val_loss: 1.0377
Epoch 254/1000
1/5 0s 42ms/step -
accuracy: 0.4141 - loss: 1.5704
Epoch 254: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4531 - loss: 1.4746 - val_accuracy: 0.6766 - val_loss: 1.0386
Epoch 255/1000
1/5 0s 46ms/step -
accuracy: 0.4688 - loss: 1.5121
Epoch 255: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4801 - loss: 1.4913 - val_accuracy: 0.6716 - val_loss: 1.0397
Epoch 256/1000
1/5 0s 36ms/step -
accuracy: 0.3984 - loss: 1.6275
Epoch 256: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4660 - loss: 1.5011 - val_accuracy: 0.6766 - val_loss: 1.0367
Epoch 257/1000
1/5 0s 34ms/step -
accuracy: 0.4766 - loss: 1.5476

Epoch 257: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5026 - loss: 1.4719 - val_accuracy: 0.6766 - val_loss: 1.0340
Epoch 258/1000
1/5 0s 31ms/step -
accuracy: 0.4688 - loss: 1.4416
Epoch 258: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4666 - loss: 1.4916 - val_accuracy: 0.6816 - val_loss: 1.0315
Epoch 259/1000
1/5 0s 33ms/step -
accuracy: 0.5156 - loss: 1.4791
Epoch 259: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5021 - loss: 1.4616 - val_accuracy: 0.6766 - val_loss: 1.0303
Epoch 260/1000
1/5 0s 33ms/step -
accuracy: 0.4531 - loss: 1.4512
Epoch 260: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4698 - loss: 1.5043 - val_accuracy: 0.6716 - val_loss: 1.0281
Epoch 261/1000
1/5 0s 33ms/step -
accuracy: 0.4766 - loss: 1.5448
Epoch 261: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4841 - loss: 1.4751 - val_accuracy: 0.6667 - val_loss: 1.0287
Epoch 262/1000
1/5 0s 33ms/step -
accuracy: 0.5156 - loss: 1.3909
Epoch 262: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5098 - loss: 1.4110 - val_accuracy: 0.6667 - val_loss: 1.0278
Epoch 263/1000
1/5 0s 30ms/step -
accuracy: 0.5078 - loss: 1.4407
Epoch 263: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4933 - loss: 1.4331 - val_accuracy: 0.6667 - val_loss: 1.0252
Epoch 264/1000
1/5 0s 42ms/step -
accuracy: 0.4922 - loss: 1.4644
Epoch 264: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5030 - loss: 1.4409 - val_accuracy: 0.6716 - val_loss: 1.0220
Epoch 265/1000
1/5 0s 37ms/step -
accuracy: 0.4766 - loss: 1.4970

Epoch 265: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4809 - loss: 1.4751 - val_accuracy: 0.6816 - val_loss: 1.0198
Epoch 266/1000
1/5 0s 43ms/step -
accuracy: 0.4609 - loss: 1.3964
Epoch 266: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4991 - loss: 1.4446 - val_accuracy: 0.6866 - val_loss: 1.0162
Epoch 267/1000
1/5 0s 34ms/step -
accuracy: 0.4766 - loss: 1.5730
Epoch 267: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4851 - loss: 1.4960 - val_accuracy: 0.6866 - val_loss: 1.0144
Epoch 268/1000
1/5 0s 35ms/step -
accuracy: 0.5391 - loss: 1.4676
Epoch 268: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5021 - loss: 1.5137 - val_accuracy: 0.6816 - val_loss: 1.0118
Epoch 269/1000
1/5 0s 43ms/step -
accuracy: 0.5938 - loss: 1.3574
Epoch 269: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.5204 - loss: 1.4159 - val_accuracy: 0.6816 - val_loss: 1.0094
Epoch 270/1000
1/5 0s 39ms/step -
accuracy: 0.5312 - loss: 1.3552
Epoch 270: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5159 - loss: 1.4070 - val_accuracy: 0.6866 - val_loss: 1.0092
Epoch 271/1000
1/5 0s 36ms/step -
accuracy: 0.4766 - loss: 1.3903
Epoch 271: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4939 - loss: 1.3896 - val_accuracy: 0.7065 - val_loss: 1.0054
Epoch 272/1000
1/5 0s 34ms/step -
accuracy: 0.4922 - loss: 1.5059
Epoch 272: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5002 - loss: 1.4251 - val_accuracy: 0.6816 - val_loss: 1.0009
Epoch 273/1000
1/5 0s 78ms/step -
accuracy: 0.5703 - loss: 1.3372

Epoch 273: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5240 - loss: 1.3884 - val_accuracy: 0.6667 - val_loss: 1.0011
Epoch 274/1000
1/5 0s 34ms/step -
accuracy: 0.4453 - loss: 1.4756
Epoch 274: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4743 - loss: 1.4690 - val_accuracy: 0.6617 - val_loss: 1.0009
Epoch 275/1000
1/5 0s 35ms/step -
accuracy: 0.4141 - loss: 1.6067
Epoch 275: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4558 - loss: 1.5284 - val_accuracy: 0.6617 - val_loss: 1.0041
Epoch 276/1000
1/5 0s 33ms/step -
accuracy: 0.5469 - loss: 1.3816
Epoch 276: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4944 - loss: 1.4558 - val_accuracy: 0.6716 - val_loss: 1.0046
Epoch 277/1000
1/5 0s 38ms/step -
accuracy: 0.4531 - loss: 1.4339
Epoch 277: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4789 - loss: 1.4528 - val_accuracy: 0.6716 - val_loss: 1.0053
Epoch 278/1000
1/5 0s 32ms/step -
accuracy: 0.5156 - loss: 1.2995
Epoch 278: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5192 - loss: 1.3394 - val_accuracy: 0.6816 - val_loss: 1.0017
Epoch 279/1000
1/5 0s 44ms/step -
accuracy: 0.4922 - loss: 1.4954
Epoch 279: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4883 - loss: 1.4849 - val_accuracy: 0.6866 - val_loss: 0.9999
Epoch 280/1000
1/5 0s 36ms/step -
accuracy: 0.4375 - loss: 1.5452
Epoch 280: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4883 - loss: 1.4393 - val_accuracy: 0.6866 - val_loss: 0.9996
Epoch 281/1000
1/5 0s 34ms/step -
accuracy: 0.5156 - loss: 1.4157

Epoch 281: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5092 - loss: 1.4210 - val_accuracy: 0.6816 - val_loss: 0.9991
Epoch 282/1000
1/5 0s 39ms/step -
accuracy: 0.5000 - loss: 1.5057
Epoch 282: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4892 - loss: 1.4770 - val_accuracy: 0.6816 - val_loss: 0.9970
Epoch 283/1000
1/5 0s 33ms/step -
accuracy: 0.4766 - loss: 1.5110
Epoch 283: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4896 - loss: 1.4630 - val_accuracy: 0.6816 - val_loss: 0.9960
Epoch 284/1000
1/5 0s 37ms/step -
accuracy: 0.4375 - loss: 1.5534
Epoch 284: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.4671 - loss: 1.4812 - val_accuracy: 0.6866 - val_loss: 0.9909
Epoch 285/1000
1/5 0s 33ms/step -
accuracy: 0.4844 - loss: 1.5080
Epoch 285: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4969 - loss: 1.4258 - val_accuracy: 0.6816 - val_loss: 0.9877
Epoch 286/1000
1/5 0s 33ms/step -
accuracy: 0.5156 - loss: 1.3573
Epoch 286: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4902 - loss: 1.4419 - val_accuracy: 0.6816 - val_loss: 0.9861
Epoch 287/1000
1/5 0s 35ms/step -
accuracy: 0.4609 - loss: 1.4045
Epoch 287: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4794 - loss: 1.4489 - val_accuracy: 0.6766 - val_loss: 0.9881
Epoch 288/1000
1/5 0s 32ms/step -
accuracy: 0.5703 - loss: 1.3410
Epoch 288: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5225 - loss: 1.3819 - val_accuracy: 0.6766 - val_loss: 0.9867
Epoch 289/1000
1/5 0s 34ms/step -
accuracy: 0.5391 - loss: 1.3446

Epoch 289: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5310 - loss: 1.3853 - val_accuracy: 0.6716 - val_loss: 0.9852
Epoch 290/1000
1/5 0s 41ms/step -
accuracy: 0.4453 - loss: 1.4314
Epoch 290: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4828 - loss: 1.4249 - val_accuracy: 0.6667 - val_loss: 0.9870
Epoch 291/1000
1/5 0s 34ms/step -
accuracy: 0.5000 - loss: 1.4862
Epoch 291: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5020 - loss: 1.4195 - val_accuracy: 0.6965 - val_loss: 0.9859
Epoch 292/1000
1/5 0s 38ms/step -
accuracy: 0.5234 - loss: 1.2950
Epoch 292: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.4931 - loss: 1.3854 - val_accuracy: 0.6766 - val_loss: 0.9817
Epoch 293/1000
1/5 0s 35ms/step -
accuracy: 0.4844 - loss: 1.5249
Epoch 293: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5028 - loss: 1.4430 - val_accuracy: 0.6816 - val_loss: 0.9738
Epoch 294/1000
1/5 0s 31ms/step -
accuracy: 0.5000 - loss: 1.4455
Epoch 294: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.4861 - loss: 1.4630 - val_accuracy: 0.6766 - val_loss: 0.9734
Epoch 295/1000
1/5 0s 34ms/step -
accuracy: 0.5312 - loss: 1.3572
Epoch 295: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4980 - loss: 1.4188 - val_accuracy: 0.6766 - val_loss: 0.9766
Epoch 296/1000
1/5 0s 34ms/step -
accuracy: 0.5156 - loss: 1.3195
Epoch 296: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5080 - loss: 1.3695 - val_accuracy: 0.6766 - val_loss: 0.9762
Epoch 297/1000
1/5 0s 40ms/step -
accuracy: 0.4609 - loss: 1.5173

Epoch 297: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4952 - loss: 1.3862 - val_accuracy: 0.6766 - val_loss: 0.9734
Epoch 298/1000
1/5 0s 37ms/step -
accuracy: 0.5547 - loss: 1.3472
Epoch 298: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5415 - loss: 1.3794 - val_accuracy: 0.6816 - val_loss: 0.9680
Epoch 299/1000
1/5 0s 31ms/step -
accuracy: 0.4844 - loss: 1.3440
Epoch 299: saving model to keypoint_classifier.keras
5/5 0s 38ms/step -
accuracy: 0.4895 - loss: 1.3850 - val_accuracy: 0.6915 - val_loss: 0.9651
Epoch 300/1000
1/5 0s 48ms/step -
accuracy: 0.5078 - loss: 1.4145
Epoch 300: saving model to keypoint_classifier.keras
5/5 0s 23ms/step -
accuracy: 0.4873 - loss: 1.4473 - val_accuracy: 0.7164 - val_loss: 0.9634
Epoch 301/1000
1/5 0s 33ms/step -
accuracy: 0.4531 - loss: 1.4717
Epoch 301: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.4877 - loss: 1.4597 - val_accuracy: 0.6766 - val_loss: 0.9663
Epoch 302/1000
1/5 0s 38ms/step -
accuracy: 0.5391 - loss: 1.4266
Epoch 302: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5122 - loss: 1.4358 - val_accuracy: 0.6716 - val_loss: 0.9727
Epoch 303/1000
1/5 0s 34ms/step -
accuracy: 0.4609 - loss: 1.3717
Epoch 303: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4960 - loss: 1.4018 - val_accuracy: 0.6766 - val_loss: 0.9742
Epoch 304/1000
1/5 0s 36ms/step -
accuracy: 0.4688 - loss: 1.4728
Epoch 304: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4487 - loss: 1.5149 - val_accuracy: 0.6866 - val_loss: 0.9741
Epoch 305/1000
1/5 0s 33ms/step -
accuracy: 0.4922 - loss: 1.3957

Epoch 305: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4900 - loss: 1.3860 - val_accuracy: 0.6816 - val_loss: 0.9708
Epoch 306/1000
1/5 0s 36ms/step -
accuracy: 0.5078 - loss: 1.4391
Epoch 306: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4943 - loss: 1.4599 - val_accuracy: 0.6816 - val_loss: 0.9696
Epoch 307/1000
1/5 0s 38ms/step -
accuracy: 0.5469 - loss: 1.3199
Epoch 307: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5117 - loss: 1.4161 - val_accuracy: 0.6816 - val_loss: 0.9678
Epoch 308/1000
1/5 0s 35ms/step -
accuracy: 0.6406 - loss: 1.2093
Epoch 308: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5497 - loss: 1.3504 - val_accuracy: 0.6866 - val_loss: 0.9650
Epoch 309/1000
1/5 0s 35ms/step -
accuracy: 0.5156 - loss: 1.4712
Epoch 309: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5188 - loss: 1.3938 - val_accuracy: 0.6816 - val_loss: 0.9657
Epoch 310/1000
1/5 0s 37ms/step -
accuracy: 0.5391 - loss: 1.3666
Epoch 310: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5422 - loss: 1.3840 - val_accuracy: 0.6716 - val_loss: 0.9658
Epoch 311/1000
1/5 0s 33ms/step -
accuracy: 0.4609 - loss: 1.5141
Epoch 311: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4717 - loss: 1.4755 - val_accuracy: 0.6766 - val_loss: 0.9639
Epoch 312/1000
1/5 0s 34ms/step -
accuracy: 0.4922 - loss: 1.4322
Epoch 312: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5164 - loss: 1.3841 - val_accuracy: 0.6766 - val_loss: 0.9648
Epoch 313/1000
1/5 0s 34ms/step -
accuracy: 0.4688 - loss: 1.4601

Epoch 313: saving model to keypoint_classifier.keras
5/5 0s 38ms/step -
accuracy: 0.5051 - loss: 1.4082 - val_accuracy: 0.6766 - val_loss: 0.9652
Epoch 314/1000
1/5 0s 36ms/step -
accuracy: 0.5078 - loss: 1.4230
Epoch 314: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5084 - loss: 1.3909 - val_accuracy: 0.6716 - val_loss: 0.9644
Epoch 315/1000
1/5 0s 46ms/step -
accuracy: 0.6172 - loss: 1.2964
Epoch 315: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5270 - loss: 1.3742 - val_accuracy: 0.6716 - val_loss: 0.9619
Epoch 316/1000
1/5 0s 33ms/step -
accuracy: 0.5625 - loss: 1.2139
Epoch 316: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5361 - loss: 1.3116 - val_accuracy: 0.6766 - val_loss: 0.9592
Epoch 317/1000
1/5 0s 35ms/step -
accuracy: 0.5234 - loss: 1.3060
Epoch 317: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5126 - loss: 1.3785 - val_accuracy: 0.6866 - val_loss: 0.9572
Epoch 318/1000
1/5 0s 38ms/step -
accuracy: 0.4922 - loss: 1.4454
Epoch 318: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4800 - loss: 1.4290 - val_accuracy: 0.6915 - val_loss: 0.9528
Epoch 319/1000
1/5 0s 35ms/step -
accuracy: 0.4922 - loss: 1.4123
Epoch 319: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5077 - loss: 1.4129 - val_accuracy: 0.6816 - val_loss: 0.9512
Epoch 320/1000
1/5 0s 36ms/step -
accuracy: 0.5000 - loss: 1.3321
Epoch 320: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5316 - loss: 1.3258 - val_accuracy: 0.6915 - val_loss: 0.9476
Epoch 321/1000
1/5 0s 30ms/step -
accuracy: 0.4609 - loss: 1.3734

Epoch 321: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5089 - loss: 1.3590 - val_accuracy: 0.6915 - val_loss: 0.9450
Epoch 322/1000
1/5 0s 39ms/step -
accuracy: 0.5547 - loss: 1.3281
Epoch 322: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5127 - loss: 1.3177 - val_accuracy: 0.6915 - val_loss: 0.9433
Epoch 323/1000
1/5 0s 32ms/step -
accuracy: 0.5859 - loss: 1.3257
Epoch 323: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5292 - loss: 1.3761 - val_accuracy: 0.6915 - val_loss: 0.9414
Epoch 324/1000
1/5 0s 35ms/step -
accuracy: 0.4766 - loss: 1.4276
Epoch 324: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4715 - loss: 1.4489 - val_accuracy: 0.6866 - val_loss: 0.9428
Epoch 325/1000
1/5 0s 36ms/step -
accuracy: 0.5859 - loss: 1.3492
Epoch 325: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5188 - loss: 1.3981 - val_accuracy: 0.6866 - val_loss: 0.9456
Epoch 326/1000
1/5 0s 40ms/step -
accuracy: 0.4766 - loss: 1.4166
Epoch 326: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4950 - loss: 1.3981 - val_accuracy: 0.7015 - val_loss: 0.9459
Epoch 327/1000
1/5 0s 63ms/step -
accuracy: 0.5156 - loss: 1.4245
Epoch 327: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5134 - loss: 1.4171 - val_accuracy: 0.6816 - val_loss: 0.9421
Epoch 328/1000
1/5 0s 41ms/step -
accuracy: 0.5000 - loss: 1.3990
Epoch 328: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4965 - loss: 1.3852 - val_accuracy: 0.6766 - val_loss: 0.9385
Epoch 329/1000
1/5 0s 41ms/step -
accuracy: 0.5156 - loss: 1.4788

Epoch 329: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5140 - loss: 1.4360 - val_accuracy: 0.6716 - val_loss: 0.9417
Epoch 330/1000
1/5 0s 35ms/step -
accuracy: 0.5000 - loss: 1.3927
Epoch 330: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5299 - loss: 1.3420 - val_accuracy: 0.6866 - val_loss: 0.9377
Epoch 331/1000
1/5 0s 36ms/step -
accuracy: 0.5625 - loss: 1.3538
Epoch 331: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5305 - loss: 1.3630 - val_accuracy: 0.6965 - val_loss: 0.9356
Epoch 332/1000
1/5 0s 34ms/step -
accuracy: 0.5391 - loss: 1.4330
Epoch 332: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5293 - loss: 1.4163 - val_accuracy: 0.6965 - val_loss: 0.9318
Epoch 333/1000
1/5 0s 33ms/step -
accuracy: 0.5703 - loss: 1.2459
Epoch 333: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5439 - loss: 1.3068 - val_accuracy: 0.6965 - val_loss: 0.9305
Epoch 334/1000
1/5 0s 29ms/step -
accuracy: 0.5000 - loss: 1.3959
Epoch 334: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.4994 - loss: 1.4189 - val_accuracy: 0.6915 - val_loss: 0.9285
Epoch 335/1000
1/5 0s 40ms/step -
accuracy: 0.5156 - loss: 1.3309
Epoch 335: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5026 - loss: 1.3745 - val_accuracy: 0.6816 - val_loss: 0.9254
Epoch 336/1000
1/5 0s 45ms/step -
accuracy: 0.5000 - loss: 1.5101
Epoch 336: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5115 - loss: 1.4575 - val_accuracy: 0.6766 - val_loss: 0.9254
Epoch 337/1000
1/5 0s 38ms/step -
accuracy: 0.5234 - loss: 1.3630

Epoch 337: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5076 - loss: 1.4005 - val_accuracy: 0.6816 - val_loss: 0.9258
Epoch 338/1000
1/5 0s 31ms/step -
accuracy: 0.4141 - loss: 1.4412
Epoch 338: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4771 - loss: 1.3975 - val_accuracy: 0.6866 - val_loss: 0.9252
Epoch 339/1000
1/5 0s 41ms/step -
accuracy: 0.6094 - loss: 1.2733
Epoch 339: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.5609 - loss: 1.3401 - val_accuracy: 0.6866 - val_loss: 0.9266
Epoch 340/1000
1/5 0s 35ms/step -
accuracy: 0.5078 - loss: 1.4526
Epoch 340: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4965 - loss: 1.4357 - val_accuracy: 0.6716 - val_loss: 0.9266
Epoch 341/1000
1/5 0s 35ms/step -
accuracy: 0.4531 - loss: 1.4708
Epoch 341: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5020 - loss: 1.3947 - val_accuracy: 0.6816 - val_loss: 0.9269
Epoch 342/1000
1/5 0s 35ms/step -
accuracy: 0.5078 - loss: 1.4379
Epoch 342: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5224 - loss: 1.3854 - val_accuracy: 0.6866 - val_loss: 0.9234
Epoch 343/1000
1/5 0s 32ms/step -
accuracy: 0.5156 - loss: 1.4737
Epoch 343: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5173 - loss: 1.3943 - val_accuracy: 0.6915 - val_loss: 0.9200
Epoch 344/1000
1/5 0s 36ms/step -
accuracy: 0.5625 - loss: 1.3223
Epoch 344: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5256 - loss: 1.3890 - val_accuracy: 0.6965 - val_loss: 0.9179
Epoch 345/1000
1/5 0s 32ms/step -
accuracy: 0.5938 - loss: 1.2673

Epoch 345: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5450 - loss: 1.3347 - val_accuracy: 0.6915 - val_loss: 0.9156
Epoch 346/1000
1/5 0s 34ms/step -
accuracy: 0.5469 - loss: 1.4194
Epoch 346: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5352 - loss: 1.3917 - val_accuracy: 0.6915 - val_loss: 0.9162
Epoch 347/1000
1/5 0s 33ms/step -
accuracy: 0.5938 - loss: 1.2516
Epoch 347: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5590 - loss: 1.3356 - val_accuracy: 0.6816 - val_loss: 0.9155
Epoch 348/1000
1/5 0s 35ms/step -
accuracy: 0.5000 - loss: 1.3561
Epoch 348: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4963 - loss: 1.3858 - val_accuracy: 0.6816 - val_loss: 0.9152
Epoch 349/1000
1/5 0s 40ms/step -
accuracy: 0.5234 - loss: 1.4319
Epoch 349: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5459 - loss: 1.3820 - val_accuracy: 0.6816 - val_loss: 0.9157
Epoch 350/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.2825
Epoch 350: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5317 - loss: 1.3333 - val_accuracy: 0.6866 - val_loss: 0.9185
Epoch 351/1000
1/5 0s 32ms/step -
accuracy: 0.5469 - loss: 1.3128
Epoch 351: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5143 - loss: 1.3796 - val_accuracy: 0.6915 - val_loss: 0.9206
Epoch 352/1000
1/5 0s 37ms/step -
accuracy: 0.5547 - loss: 1.3393
Epoch 352: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.5398 - loss: 1.3407 - val_accuracy: 0.6965 - val_loss: 0.9192
Epoch 353/1000
1/5 0s 38ms/step -
accuracy: 0.5078 - loss: 1.5110

Epoch 353: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5142 - loss: 1.4277 - val_accuracy: 0.6866 - val_loss: 0.9201
Epoch 354/1000
1/5 0s 28ms/step -
accuracy: 0.5469 - loss: 1.3401
Epoch 354: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5377 - loss: 1.3220 - val_accuracy: 0.6965 - val_loss: 0.9167
Epoch 355/1000
1/5 0s 33ms/step -
accuracy: 0.4688 - loss: 1.3875
Epoch 355: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4803 - loss: 1.3931 - val_accuracy: 0.6965 - val_loss: 0.9135
Epoch 356/1000
1/5 0s 33ms/step -
accuracy: 0.5703 - loss: 1.3218
Epoch 356: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5401 - loss: 1.3314 - val_accuracy: 0.7015 - val_loss: 0.9125
Epoch 357/1000
1/5 0s 31ms/step -
accuracy: 0.5469 - loss: 1.2553
Epoch 357: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5162 - loss: 1.2927 - val_accuracy: 0.7164 - val_loss: 0.9091
Epoch 358/1000
1/5 0s 27ms/step -
accuracy: 0.5234 - loss: 1.4224
Epoch 358: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5006 - loss: 1.4145 - val_accuracy: 0.7065 - val_loss: 0.9087
Epoch 359/1000
1/5 0s 34ms/step -
accuracy: 0.4688 - loss: 1.4947
Epoch 359: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5131 - loss: 1.4049 - val_accuracy: 0.6915 - val_loss: 0.9104
Epoch 360/1000
1/5 0s 49ms/step -
accuracy: 0.4453 - loss: 1.4665
Epoch 360: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4933 - loss: 1.3722 - val_accuracy: 0.6866 - val_loss: 0.9108
Epoch 361/1000
1/5 0s 34ms/step -
accuracy: 0.5234 - loss: 1.3969

Epoch 361: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5132 - loss: 1.3811 - val_accuracy: 0.6915 - val_loss: 0.9103
Epoch 362/1000
1/5 0s 32ms/step -
accuracy: 0.5469 - loss: 1.2847
Epoch 362: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5070 - loss: 1.3492 - val_accuracy: 0.6915 - val_loss: 0.9129
Epoch 363/1000
1/5 0s 35ms/step -
accuracy: 0.5000 - loss: 1.3864
Epoch 363: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5158 - loss: 1.3528 - val_accuracy: 0.6866 - val_loss: 0.9098
Epoch 364/1000
1/5 0s 34ms/step -
accuracy: 0.4453 - loss: 1.4955
Epoch 364: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.4840 - loss: 1.4295 - val_accuracy: 0.6816 - val_loss: 0.9114
Epoch 365/1000
1/5 0s 33ms/step -
accuracy: 0.5781 - loss: 1.2124
Epoch 365: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5291 - loss: 1.3444 - val_accuracy: 0.6816 - val_loss: 0.9147
Epoch 366/1000
1/5 0s 36ms/step -
accuracy: 0.5234 - loss: 1.5059
Epoch 366: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5191 - loss: 1.3813 - val_accuracy: 0.6816 - val_loss: 0.9140
Epoch 367/1000
1/5 0s 30ms/step -
accuracy: 0.4844 - loss: 1.3657
Epoch 367: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5220 - loss: 1.3583 - val_accuracy: 0.6766 - val_loss: 0.9108
Epoch 368/1000
1/5 0s 36ms/step -
accuracy: 0.5312 - loss: 1.4294
Epoch 368: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5328 - loss: 1.3743 - val_accuracy: 0.6866 - val_loss: 0.9033
Epoch 369/1000
1/5 0s 36ms/step -
accuracy: 0.5156 - loss: 1.3582

Epoch 369: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5165 - loss: 1.3825 - val_accuracy: 0.6816 - val_loss: 0.9001
Epoch 370/1000
1/5 0s 35ms/step -
accuracy: 0.5000 - loss: 1.4641
Epoch 370: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4907 - loss: 1.4432 - val_accuracy: 0.6866 - val_loss: 0.8999
Epoch 371/1000
1/5 0s 39ms/step -
accuracy: 0.4922 - loss: 1.4237
Epoch 371: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5009 - loss: 1.3879 - val_accuracy: 0.6915 - val_loss: 0.9033
Epoch 372/1000
1/5 0s 33ms/step -
accuracy: 0.5078 - loss: 1.4085
Epoch 372: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5132 - loss: 1.3580 - val_accuracy: 0.6965 - val_loss: 0.9057
Epoch 373/1000
1/5 0s 33ms/step -
accuracy: 0.4766 - loss: 1.4113
Epoch 373: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4998 - loss: 1.3882 - val_accuracy: 0.7065 - val_loss: 0.9041
Epoch 374/1000
1/5 0s 34ms/step -
accuracy: 0.5703 - loss: 1.1809
Epoch 374: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5316 - loss: 1.3056 - val_accuracy: 0.7015 - val_loss: 0.9032
Epoch 375/1000
1/5 0s 40ms/step -
accuracy: 0.4297 - loss: 1.3933
Epoch 375: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4929 - loss: 1.3650 - val_accuracy: 0.6965 - val_loss: 0.9004
Epoch 376/1000
1/5 0s 38ms/step -
accuracy: 0.5156 - loss: 1.4460
Epoch 376: saving model to keypoint_classifier.keras
5/5 0s 41ms/step -
accuracy: 0.5313 - loss: 1.3799 - val_accuracy: 0.6915 - val_loss: 0.8938
Epoch 377/1000
1/5 0s 34ms/step -
accuracy: 0.5234 - loss: 1.4562

Epoch 377: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5382 - loss: 1.3654 - val_accuracy: 0.6965 - val_loss: 0.8880
Epoch 378/1000
1/5 0s 35ms/step -
accuracy: 0.4844 - loss: 1.3221
Epoch 378: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5034 - loss: 1.3576 - val_accuracy: 0.6915 - val_loss: 0.8858
Epoch 379/1000
1/5 0s 33ms/step -
accuracy: 0.5547 - loss: 1.2992
Epoch 379: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5668 - loss: 1.2762 - val_accuracy: 0.6915 - val_loss: 0.8863
Epoch 380/1000
1/5 0s 31ms/step -
accuracy: 0.5312 - loss: 1.3676
Epoch 380: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5205 - loss: 1.3463 - val_accuracy: 0.6866 - val_loss: 0.8853
Epoch 381/1000
1/5 0s 34ms/step -
accuracy: 0.4688 - loss: 1.3675
Epoch 381: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5175 - loss: 1.3466 - val_accuracy: 0.6816 - val_loss: 0.8861
Epoch 382/1000
1/5 0s 34ms/step -
accuracy: 0.5000 - loss: 1.3629
Epoch 382: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5299 - loss: 1.3049 - val_accuracy: 0.6866 - val_loss: 0.8853
Epoch 383/1000
1/5 0s 34ms/step -
accuracy: 0.6094 - loss: 1.2882
Epoch 383: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5564 - loss: 1.3099 - val_accuracy: 0.6915 - val_loss: 0.8837
Epoch 384/1000
1/5 0s 39ms/step -
accuracy: 0.5703 - loss: 1.3631
Epoch 384: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5766 - loss: 1.3147 - val_accuracy: 0.7015 - val_loss: 0.8812
Epoch 385/1000
1/5 0s 36ms/step -
accuracy: 0.4844 - loss: 1.3983

Epoch 385: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5246 - loss: 1.3420 - val_accuracy: 0.7114 - val_loss: 0.8822
Epoch 386/1000
1/5 0s 31ms/step -
accuracy: 0.5078 - loss: 1.3985
Epoch 386: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5218 - loss: 1.3638 - val_accuracy: 0.7114 - val_loss: 0.8854
Epoch 387/1000
1/5 0s 33ms/step -
accuracy: 0.5000 - loss: 1.2774
Epoch 387: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5029 - loss: 1.2912 - val_accuracy: 0.7065 - val_loss: 0.8856
Epoch 388/1000
1/5 0s 70ms/step -
accuracy: 0.5078 - loss: 1.3624
Epoch 388: saving model to keypoint_classifier.keras
5/5 0s 32ms/step -
accuracy: 0.5334 - loss: 1.3231 - val_accuracy: 0.7065 - val_loss: 0.8876
Epoch 389/1000
1/5 0s 34ms/step -
accuracy: 0.5547 - loss: 1.1993
Epoch 389: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5520 - loss: 1.2847 - val_accuracy: 0.7114 - val_loss: 0.8863
Epoch 390/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.2981
Epoch 390: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5170 - loss: 1.3514 - val_accuracy: 0.7214 - val_loss: 0.8844
Epoch 391/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.3089
Epoch 391: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5134 - loss: 1.3046 - val_accuracy: 0.7164 - val_loss: 0.8813
Epoch 392/1000
1/5 0s 38ms/step -
accuracy: 0.5312 - loss: 1.2685
Epoch 392: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5241 - loss: 1.3393 - val_accuracy: 0.7065 - val_loss: 0.8806
Epoch 393/1000
1/5 0s 33ms/step -
accuracy: 0.6016 - loss: 1.2447

Epoch 393: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5547 - loss: 1.3161 - val_accuracy: 0.7015 - val_loss: 0.8780
Epoch 394/1000
1/5 0s 41ms/step -
accuracy: 0.4844 - loss: 1.3455
Epoch 394: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5205 - loss: 1.3498 - val_accuracy: 0.7065 - val_loss: 0.8792
Epoch 395/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.3119
Epoch 395: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5418 - loss: 1.3066 - val_accuracy: 0.6965 - val_loss: 0.8816
Epoch 396/1000
1/5 0s 35ms/step -
accuracy: 0.6406 - loss: 1.1752
Epoch 396: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5715 - loss: 1.2677 - val_accuracy: 0.6965 - val_loss: 0.8800
Epoch 397/1000
1/5 0s 34ms/step -
accuracy: 0.6172 - loss: 1.3007
Epoch 397: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5756 - loss: 1.2560 - val_accuracy: 0.7065 - val_loss: 0.8782
Epoch 398/1000
1/5 0s 38ms/step -
accuracy: 0.5312 - loss: 1.4429
Epoch 398: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5616 - loss: 1.2981 - val_accuracy: 0.6915 - val_loss: 0.8722
Epoch 399/1000
1/5 0s 27ms/step -
accuracy: 0.5000 - loss: 1.3771
Epoch 399: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.5370 - loss: 1.3498 - val_accuracy: 0.6965 - val_loss: 0.8682
Epoch 400/1000
1/5 0s 35ms/step -
accuracy: 0.5859 - loss: 1.2414
Epoch 400: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5457 - loss: 1.2935 - val_accuracy: 0.6965 - val_loss: 0.8662
Epoch 401/1000
1/5 0s 35ms/step -
accuracy: 0.5078 - loss: 1.3871

Epoch 401: saving model to keypoint_classifier.keras
5/5 0s 31ms/step -
accuracy: 0.5237 - loss: 1.3196 - val_accuracy: 0.6965 - val_loss: 0.8679
Epoch 402/1000
1/5 0s 36ms/step -
accuracy: 0.5469 - loss: 1.2418
Epoch 402: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5378 - loss: 1.3188 - val_accuracy: 0.6866 - val_loss: 0.8734
Epoch 403/1000
1/5 0s 33ms/step -
accuracy: 0.4922 - loss: 1.3793
Epoch 403: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5072 - loss: 1.3306 - val_accuracy: 0.6866 - val_loss: 0.8761
Epoch 404/1000
1/5 0s 38ms/step -
accuracy: 0.5625 - loss: 1.1844
Epoch 404: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5546 - loss: 1.2416 - val_accuracy: 0.7015 - val_loss: 0.8737
Epoch 405/1000
1/5 0s 32ms/step -
accuracy: 0.5312 - loss: 1.3307
Epoch 405: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5239 - loss: 1.3468 - val_accuracy: 0.7114 - val_loss: 0.8734
Epoch 406/1000
1/5 0s 36ms/step -
accuracy: 0.4609 - loss: 1.3868
Epoch 406: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5136 - loss: 1.3606 - val_accuracy: 0.7114 - val_loss: 0.8657
Epoch 407/1000
1/5 0s 33ms/step -
accuracy: 0.5078 - loss: 1.3214
Epoch 407: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5017 - loss: 1.3263 - val_accuracy: 0.7114 - val_loss: 0.8584
Epoch 408/1000
1/5 0s 41ms/step -
accuracy: 0.4766 - loss: 1.2993
Epoch 408: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5243 - loss: 1.2927 - val_accuracy: 0.7114 - val_loss: 0.8536
Epoch 409/1000
1/5 0s 34ms/step -
accuracy: 0.6562 - loss: 1.0857

Epoch 409: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5792 - loss: 1.2120 - val_accuracy: 0.7114 - val_loss: 0.8537
Epoch 410/1000
1/5 0s 36ms/step -
accuracy: 0.5859 - loss: 1.2706
Epoch 410: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.5411 - loss: 1.3473 - val_accuracy: 0.7065 - val_loss: 0.8546
Epoch 411/1000
1/5 0s 42ms/step -
accuracy: 0.5547 - loss: 1.2517
Epoch 411: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5476 - loss: 1.3259 - val_accuracy: 0.7015 - val_loss: 0.8553
Epoch 412/1000
1/5 0s 29ms/step -
accuracy: 0.5547 - loss: 1.2807
Epoch 412: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5701 - loss: 1.2570 - val_accuracy: 0.7015 - val_loss: 0.8582
Epoch 413/1000
1/5 0s 33ms/step -
accuracy: 0.5859 - loss: 1.1635
Epoch 413: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5513 - loss: 1.2958 - val_accuracy: 0.7015 - val_loss: 0.8564
Epoch 414/1000
1/5 0s 32ms/step -
accuracy: 0.6406 - loss: 1.2251
Epoch 414: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5952 - loss: 1.2643 - val_accuracy: 0.7114 - val_loss: 0.8557
Epoch 415/1000
1/5 0s 31ms/step -
accuracy: 0.5781 - loss: 1.4147
Epoch 415: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5634 - loss: 1.3332 - val_accuracy: 0.7065 - val_loss: 0.8569
Epoch 416/1000
1/5 0s 31ms/step -
accuracy: 0.5703 - loss: 1.3326
Epoch 416: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5103 - loss: 1.3843 - val_accuracy: 0.7015 - val_loss: 0.8609
Epoch 417/1000
1/5 0s 37ms/step -
accuracy: 0.5000 - loss: 1.3390

Epoch 417: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5183 - loss: 1.2909 - val_accuracy: 0.7015 - val_loss: 0.8598
Epoch 418/1000
1/5 0s 31ms/step -
accuracy: 0.5234 - loss: 1.3064
Epoch 418: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5190 - loss: 1.3587 - val_accuracy: 0.7214 - val_loss: 0.8631
Epoch 419/1000
1/5 0s 32ms/step -
accuracy: 0.4688 - loss: 1.3536
Epoch 419: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.4977 - loss: 1.3494 - val_accuracy: 0.7313 - val_loss: 0.8619
Epoch 420/1000
1/5 0s 36ms/step -
accuracy: 0.4844 - loss: 1.4392
Epoch 420: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5415 - loss: 1.3018 - val_accuracy: 0.7164 - val_loss: 0.8587
Epoch 421/1000
1/5 0s 35ms/step -
accuracy: 0.5859 - loss: 1.3176
Epoch 421: saving model to keypoint_classifier.keras
5/5 0s 40ms/step -
accuracy: 0.5695 - loss: 1.2950 - val_accuracy: 0.7214 - val_loss: 0.8551
Epoch 422/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.4349
Epoch 422: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5350 - loss: 1.3480 - val_accuracy: 0.7114 - val_loss: 0.8540
Epoch 423/1000
1/5 0s 32ms/step -
accuracy: 0.5312 - loss: 1.3242
Epoch 423: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5327 - loss: 1.3395 - val_accuracy: 0.7065 - val_loss: 0.8499
Epoch 424/1000
1/5 0s 42ms/step -
accuracy: 0.5078 - loss: 1.3947
Epoch 424: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5366 - loss: 1.3243 - val_accuracy: 0.7114 - val_loss: 0.8507
Epoch 425/1000
1/5 0s 34ms/step -
accuracy: 0.5781 - loss: 1.2122

Epoch 425: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5436 - loss: 1.2916 - val_accuracy: 0.7065 - val_loss: 0.8541
Epoch 426/1000
1/5 0s 33ms/step -
accuracy: 0.5625 - loss: 1.2019
Epoch 426: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5298 - loss: 1.2610 - val_accuracy: 0.7015 - val_loss: 0.8546
Epoch 427/1000
1/5 0s 38ms/step -
accuracy: 0.4609 - loss: 1.4419
Epoch 427: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5035 - loss: 1.3678 - val_accuracy: 0.7015 - val_loss: 0.8519
Epoch 428/1000
1/5 0s 34ms/step -
accuracy: 0.4531 - loss: 1.4368
Epoch 428: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5208 - loss: 1.3041 - val_accuracy: 0.7015 - val_loss: 0.8510
Epoch 429/1000
1/5 0s 32ms/step -
accuracy: 0.5312 - loss: 1.2417
Epoch 429: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5315 - loss: 1.2968 - val_accuracy: 0.7065 - val_loss: 0.8472
Epoch 430/1000
1/5 0s 42ms/step -
accuracy: 0.4922 - loss: 1.2988
Epoch 430: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5308 - loss: 1.2830 - val_accuracy: 0.7065 - val_loss: 0.8460
Epoch 431/1000
1/5 0s 35ms/step -
accuracy: 0.5469 - loss: 1.2906
Epoch 431: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5412 - loss: 1.2616 - val_accuracy: 0.7015 - val_loss: 0.8480
Epoch 432/1000
1/5 0s 32ms/step -
accuracy: 0.5859 - loss: 1.2426
Epoch 432: saving model to keypoint_classifier.keras
5/5 0s 41ms/step -
accuracy: 0.5701 - loss: 1.2707 - val_accuracy: 0.7015 - val_loss: 0.8463
Epoch 433/1000
1/5 0s 37ms/step -
accuracy: 0.5312 - loss: 1.2497

Epoch 433: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5503 - loss: 1.2502 - val_accuracy: 0.7065 - val_loss: 0.8408
Epoch 434/1000
1/5 0s 35ms/step -
accuracy: 0.4609 - loss: 1.5418
Epoch 434: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.4954 - loss: 1.4370 - val_accuracy: 0.7114 - val_loss: 0.8436
Epoch 435/1000
1/5 0s 33ms/step -
accuracy: 0.4688 - loss: 1.4196
Epoch 435: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5008 - loss: 1.3153 - val_accuracy: 0.7114 - val_loss: 0.8451
Epoch 436/1000
1/5 0s 34ms/step -
accuracy: 0.5078 - loss: 1.4216
Epoch 436: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5385 - loss: 1.2964 - val_accuracy: 0.7065 - val_loss: 0.8472
Epoch 437/1000
1/5 0s 37ms/step -
accuracy: 0.5391 - loss: 1.2368
Epoch 437: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5405 - loss: 1.2975 - val_accuracy: 0.7015 - val_loss: 0.8469
Epoch 438/1000
1/5 0s 41ms/step -
accuracy: 0.5625 - loss: 1.2930
Epoch 438: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5455 - loss: 1.2952 - val_accuracy: 0.7114 - val_loss: 0.8542
Epoch 439/1000
1/5 0s 31ms/step -
accuracy: 0.5156 - loss: 1.4499
Epoch 439: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5190 - loss: 1.3578 - val_accuracy: 0.6965 - val_loss: 0.8605
Epoch 440/1000
1/5 0s 33ms/step -
accuracy: 0.5781 - loss: 1.2197
Epoch 440: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5575 - loss: 1.2387 - val_accuracy: 0.6915 - val_loss: 0.8615
Epoch 441/1000
1/5 0s 39ms/step -
accuracy: 0.6016 - loss: 1.2115

Epoch 441: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.5646 - loss: 1.2561 - val_accuracy: 0.6866 - val_loss: 0.8564
Epoch 442/1000
1/5 0s 27ms/step -
accuracy: 0.6094 - loss: 1.1907
Epoch 442: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5551 - loss: 1.2569 - val_accuracy: 0.7065 - val_loss: 0.8493
Epoch 443/1000
1/5 0s 42ms/step -
accuracy: 0.5625 - loss: 1.1292
Epoch 443: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.5476 - loss: 1.2232 - val_accuracy: 0.7164 - val_loss: 0.8432
Epoch 444/1000
1/5 0s 40ms/step -
accuracy: 0.5156 - loss: 1.2441
Epoch 444: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5364 - loss: 1.2671 - val_accuracy: 0.7164 - val_loss: 0.8356
Epoch 445/1000
1/5 0s 37ms/step -
accuracy: 0.6172 - loss: 1.1785
Epoch 445: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5843 - loss: 1.2023 - val_accuracy: 0.7214 - val_loss: 0.8330
Epoch 446/1000
1/5 0s 38ms/step -
accuracy: 0.6094 - loss: 1.2306
Epoch 446: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5778 - loss: 1.2222 - val_accuracy: 0.7214 - val_loss: 0.8326
Epoch 447/1000
1/5 0s 35ms/step -
accuracy: 0.4609 - loss: 1.5618
Epoch 447: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.4857 - loss: 1.3902 - val_accuracy: 0.7114 - val_loss: 0.8334
Epoch 448/1000
1/5 0s 32ms/step -
accuracy: 0.6016 - loss: 1.2023
Epoch 448: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5567 - loss: 1.2556 - val_accuracy: 0.7065 - val_loss: 0.8370
Epoch 449/1000
1/5 0s 36ms/step -
accuracy: 0.6484 - loss: 1.1057

Epoch 449: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5953 - loss: 1.2181 - val_accuracy: 0.7065 - val_loss: 0.8373
Epoch 450/1000
1/5 0s 37ms/step -
accuracy: 0.4531 - loss: 1.3730
Epoch 450: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5073 - loss: 1.3154 - val_accuracy: 0.7114 - val_loss: 0.8359
Epoch 451/1000
1/5 0s 43ms/step -
accuracy: 0.6641 - loss: 0.9955
Epoch 451: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5876 - loss: 1.1790 - val_accuracy: 0.7114 - val_loss: 0.8365
Epoch 452/1000
1/5 0s 36ms/step -
accuracy: 0.5469 - loss: 1.2426
Epoch 452: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5376 - loss: 1.2718 - val_accuracy: 0.7164 - val_loss: 0.8367
Epoch 453/1000
1/5 0s 30ms/step -
accuracy: 0.5391 - loss: 1.3044
Epoch 453: saving model to keypoint_classifier.keras
5/5 0s 42ms/step -
accuracy: 0.5564 - loss: 1.2744 - val_accuracy: 0.7114 - val_loss: 0.8346
Epoch 454/1000
1/5 0s 36ms/step -
accuracy: 0.5391 - loss: 1.2203
Epoch 454: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5508 - loss: 1.2391 - val_accuracy: 0.7114 - val_loss: 0.8336
Epoch 455/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.3511
Epoch 455: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5542 - loss: 1.2709 - val_accuracy: 0.7065 - val_loss: 0.8308
Epoch 456/1000
1/5 0s 39ms/step -
accuracy: 0.5078 - loss: 1.4002
Epoch 456: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5217 - loss: 1.3473 - val_accuracy: 0.7114 - val_loss: 0.8297
Epoch 457/1000
1/5 0s 36ms/step -
accuracy: 0.5391 - loss: 1.2742

Epoch 457: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5535 - loss: 1.2890 - val_accuracy: 0.7114 - val_loss: 0.8267
Epoch 458/1000
1/5 0s 39ms/step -
accuracy: 0.5312 - loss: 1.4296
Epoch 458: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5637 - loss: 1.3188 - val_accuracy: 0.7065 - val_loss: 0.8308
Epoch 459/1000
1/5 0s 35ms/step -
accuracy: 0.5703 - loss: 1.2939
Epoch 459: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5606 - loss: 1.3130 - val_accuracy: 0.7015 - val_loss: 0.8321
Epoch 460/1000
1/5 0s 42ms/step -
accuracy: 0.5312 - loss: 1.4177
Epoch 460: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5394 - loss: 1.3562 - val_accuracy: 0.7015 - val_loss: 0.8344
Epoch 461/1000
1/5 0s 31ms/step -
accuracy: 0.5625 - loss: 1.2173
Epoch 461: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5409 - loss: 1.2914 - val_accuracy: 0.7015 - val_loss: 0.8393
Epoch 462/1000
1/5 0s 38ms/step -
accuracy: 0.5234 - loss: 1.3297
Epoch 462: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5197 - loss: 1.3379 - val_accuracy: 0.7065 - val_loss: 0.8406
Epoch 463/1000
1/5 0s 30ms/step -
accuracy: 0.4922 - loss: 1.2913
Epoch 463: saving model to keypoint_classifier.keras
5/5 0s 36ms/step -
accuracy: 0.5304 - loss: 1.2612 - val_accuracy: 0.7065 - val_loss: 0.8383
Epoch 464/1000
1/5 0s 39ms/step -
accuracy: 0.5625 - loss: 1.2163
Epoch 464: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5694 - loss: 1.2256 - val_accuracy: 0.7264 - val_loss: 0.8360
Epoch 465/1000
1/5 0s 33ms/step -
accuracy: 0.5000 - loss: 1.3699

Epoch 465: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5106 - loss: 1.3293 - val_accuracy: 0.7264 - val_loss: 0.8375
Epoch 466/1000
1/5 0s 36ms/step -
accuracy: 0.5234 - loss: 1.2779
Epoch 466: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5422 - loss: 1.2765 - val_accuracy: 0.7164 - val_loss: 0.8372
Epoch 467/1000
1/5 0s 35ms/step -
accuracy: 0.5156 - loss: 1.3469
Epoch 467: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5335 - loss: 1.3123 - val_accuracy: 0.7164 - val_loss: 0.8385
Epoch 468/1000
1/5 0s 43ms/step -
accuracy: 0.5625 - loss: 1.2440
Epoch 468: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5736 - loss: 1.2365 - val_accuracy: 0.7065 - val_loss: 0.8414
Epoch 469/1000
1/5 0s 31ms/step -
accuracy: 0.5781 - loss: 1.1531
Epoch 469: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5551 - loss: 1.2384 - val_accuracy: 0.7015 - val_loss: 0.8453
Epoch 470/1000
1/5 0s 34ms/step -
accuracy: 0.5391 - loss: 1.1769
Epoch 470: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5614 - loss: 1.2002 - val_accuracy: 0.7065 - val_loss: 0.8421
Epoch 471/1000
1/5 0s 34ms/step -
accuracy: 0.5703 - loss: 1.2119
Epoch 471: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5592 - loss: 1.2448 - val_accuracy: 0.7065 - val_loss: 0.8404
Epoch 472/1000
1/5 0s 29ms/step -
accuracy: 0.5859 - loss: 1.2216
Epoch 472: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5653 - loss: 1.2406 - val_accuracy: 0.7015 - val_loss: 0.8336
Epoch 473/1000
1/5 0s 34ms/step -
accuracy: 0.5312 - loss: 1.2468

Epoch 473: saving model to keypoint_classifier.keras
5/5 0s 38ms/step -
accuracy: 0.5434 - loss: 1.2978 - val_accuracy: 0.7065 - val_loss: 0.8267
Epoch 474/1000
1/5 0s 35ms/step -
accuracy: 0.6094 - loss: 1.2357
Epoch 474: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5775 - loss: 1.2618 - val_accuracy: 0.7015 - val_loss: 0.8223
Epoch 475/1000
1/5 0s 34ms/step -
accuracy: 0.5156 - loss: 1.2671
Epoch 475: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5316 - loss: 1.2692 - val_accuracy: 0.7164 - val_loss: 0.8229
Epoch 476/1000
1/5 0s 35ms/step -
accuracy: 0.5391 - loss: 1.3768
Epoch 476: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5493 - loss: 1.2995 - val_accuracy: 0.7264 - val_loss: 0.8191
Epoch 477/1000
1/5 0s 41ms/step -
accuracy: 0.5469 - loss: 1.3336
Epoch 477: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5552 - loss: 1.2633 - val_accuracy: 0.7164 - val_loss: 0.8211
Epoch 478/1000
1/5 0s 31ms/step -
accuracy: 0.5938 - loss: 1.1141
Epoch 478: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5783 - loss: 1.2010 - val_accuracy: 0.7114 - val_loss: 0.8181
Epoch 479/1000
1/5 0s 37ms/step -
accuracy: 0.5312 - loss: 1.3206
Epoch 479: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5324 - loss: 1.3181 - val_accuracy: 0.7114 - val_loss: 0.8138
Epoch 480/1000
1/5 0s 34ms/step -
accuracy: 0.5938 - loss: 1.2931
Epoch 480: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5612 - loss: 1.2747 - val_accuracy: 0.7114 - val_loss: 0.8107
Epoch 481/1000
1/5 0s 33ms/step -
accuracy: 0.5078 - loss: 1.4347

Epoch 481: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5394 - loss: 1.3389 - val_accuracy: 0.7114 - val_loss: 0.8110
Epoch 482/1000
1/5 0s 33ms/step -
accuracy: 0.5859 - loss: 1.3084
Epoch 482: saving model to keypoint_classifier.keras
5/5 0s 41ms/step -
accuracy: 0.5803 - loss: 1.2353 - val_accuracy: 0.7114 - val_loss: 0.8147
Epoch 483/1000
1/5 0s 36ms/step -
accuracy: 0.5625 - loss: 1.2910
Epoch 483: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5525 - loss: 1.2430 - val_accuracy: 0.7114 - val_loss: 0.8134
Epoch 484/1000
1/5 0s 37ms/step -
accuracy: 0.5234 - loss: 1.3393
Epoch 484: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5591 - loss: 1.2705 - val_accuracy: 0.7214 - val_loss: 0.8162
Epoch 485/1000
1/5 0s 39ms/step -
accuracy: 0.5000 - loss: 1.3949
Epoch 485: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5011 - loss: 1.3187 - val_accuracy: 0.7264 - val_loss: 0.8221
Epoch 486/1000
1/5 0s 35ms/step -
accuracy: 0.5859 - loss: 1.1831
Epoch 486: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5650 - loss: 1.2503 - val_accuracy: 0.7214 - val_loss: 0.8238
Epoch 487/1000
1/5 0s 37ms/step -
accuracy: 0.5547 - loss: 1.2449
Epoch 487: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5539 - loss: 1.2463 - val_accuracy: 0.7264 - val_loss: 0.8243
Epoch 488/1000
1/5 0s 34ms/step -
accuracy: 0.5938 - loss: 1.3013
Epoch 488: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5845 - loss: 1.2365 - val_accuracy: 0.7164 - val_loss: 0.8226
Epoch 489/1000
1/5 0s 36ms/step -
accuracy: 0.5469 - loss: 1.3247

Epoch 489: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5588 - loss: 1.2467 - val_accuracy: 0.7164 - val_loss: 0.8169
Epoch 490/1000
1/5 0s 36ms/step -
accuracy: 0.6484 - loss: 1.0791
Epoch 490: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.6009 - loss: 1.2060 - val_accuracy: 0.7214 - val_loss: 0.8127
Epoch 491/1000
1/5 0s 39ms/step -
accuracy: 0.5703 - loss: 1.3599
Epoch 491: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5812 - loss: 1.2635 - val_accuracy: 0.7363 - val_loss: 0.8108
Epoch 492/1000
1/5 0s 39ms/step -
accuracy: 0.5781 - loss: 1.1366
Epoch 492: saving model to keypoint_classifier.keras
5/5 0s 38ms/step -
accuracy: 0.5717 - loss: 1.2156 - val_accuracy: 0.7313 - val_loss: 0.8112
Epoch 493/1000
1/5 0s 41ms/step -
accuracy: 0.5312 - loss: 1.3010
Epoch 493: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5410 - loss: 1.2895 - val_accuracy: 0.7214 - val_loss: 0.8106
Epoch 494/1000
1/5 0s 33ms/step -
accuracy: 0.5938 - loss: 1.1307
Epoch 494: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5674 - loss: 1.2109 - val_accuracy: 0.7065 - val_loss: 0.8106
Epoch 495/1000
1/5 0s 33ms/step -
accuracy: 0.5781 - loss: 1.2891
Epoch 495: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5670 - loss: 1.2594 - val_accuracy: 0.7065 - val_loss: 0.8093
Epoch 496/1000
1/5 0s 36ms/step -
accuracy: 0.5625 - loss: 1.3000
Epoch 496: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5944 - loss: 1.2117 - val_accuracy: 0.7065 - val_loss: 0.8086
Epoch 497/1000
1/5 0s 33ms/step -
accuracy: 0.5547 - loss: 1.2805

Epoch 497: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5755 - loss: 1.2438 - val_accuracy: 0.7065 - val_loss: 0.8083
Epoch 498/1000
1/5 0s 36ms/step -
accuracy: 0.6094 - loss: 1.1872
Epoch 498: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.6006 - loss: 1.2014 - val_accuracy: 0.7114 - val_loss: 0.8129
Epoch 499/1000
1/5 0s 35ms/step -
accuracy: 0.5312 - loss: 1.3389
Epoch 499: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5426 - loss: 1.3122 - val_accuracy: 0.7114 - val_loss: 0.8154
Epoch 500/1000
1/5 0s 41ms/step -
accuracy: 0.5703 - loss: 1.2857
Epoch 500: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5500 - loss: 1.2885 - val_accuracy: 0.7065 - val_loss: 0.8163
Epoch 501/1000
1/5 0s 45ms/step -
accuracy: 0.6328 - loss: 1.2024
Epoch 501: saving model to keypoint_classifier.keras
5/5 0s 37ms/step -
accuracy: 0.5677 - loss: 1.2665 - val_accuracy: 0.7065 - val_loss: 0.8105
Epoch 502/1000
1/5 0s 37ms/step -
accuracy: 0.5859 - loss: 1.1984
Epoch 502: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5633 - loss: 1.2408 - val_accuracy: 0.7065 - val_loss: 0.8050
Epoch 503/1000
1/5 0s 34ms/step -
accuracy: 0.5547 - loss: 1.2057
Epoch 503: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5388 - loss: 1.2490 - val_accuracy: 0.7015 - val_loss: 0.8025
Epoch 504/1000
1/5 0s 28ms/step -
accuracy: 0.5547 - loss: 1.3143
Epoch 504: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5524 - loss: 1.2728 - val_accuracy: 0.6965 - val_loss: 0.8076
Epoch 505/1000
1/5 0s 31ms/step -
accuracy: 0.5859 - loss: 1.2145

Epoch 505: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5788 - loss: 1.2144 - val_accuracy: 0.7114 - val_loss: 0.8145
Epoch 506/1000
1/5 0s 34ms/step -
accuracy: 0.5078 - loss: 1.2280
Epoch 506: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5496 - loss: 1.2367 - val_accuracy: 0.7313 - val_loss: 0.8156
Epoch 507/1000
1/5 0s 33ms/step -
accuracy: 0.5312 - loss: 1.2636
Epoch 507: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5543 - loss: 1.2406 - val_accuracy: 0.7363 - val_loss: 0.8108
Epoch 508/1000
1/5 0s 33ms/step -
accuracy: 0.5391 - loss: 1.2120
Epoch 508: saving model to keypoint_classifier.keras
5/5 0s 25ms/step -
accuracy: 0.5495 - loss: 1.2238 - val_accuracy: 0.7264 - val_loss: 0.8057
Epoch 509/1000
1/5 0s 39ms/step -
accuracy: 0.5469 - loss: 1.3827
Epoch 509: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5515 - loss: 1.2976 - val_accuracy: 0.7214 - val_loss: 0.8062
Epoch 510/1000
1/5 0s 28ms/step -
accuracy: 0.5469 - loss: 1.2401
Epoch 510: saving model to keypoint_classifier.keras
5/5 0s 39ms/step -
accuracy: 0.5254 - loss: 1.2955 - val_accuracy: 0.7264 - val_loss: 0.8083
Epoch 511/1000
1/5 0s 36ms/step -
accuracy: 0.5703 - loss: 1.2931
Epoch 511: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5617 - loss: 1.2673 - val_accuracy: 0.7313 - val_loss: 0.8110
Epoch 512/1000
1/5 0s 43ms/step -
accuracy: 0.6250 - loss: 1.1511
Epoch 512: saving model to keypoint_classifier.keras
5/5 0s 33ms/step -
accuracy: 0.5828 - loss: 1.1663 - val_accuracy: 0.7264 - val_loss: 0.8081
Epoch 513/1000
1/5 0s 32ms/step -
accuracy: 0.5000 - loss: 1.3001

Epoch 513: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5336 - loss: 1.2407 - val_accuracy: 0.7214 - val_loss: 0.8024
Epoch 514/1000
1/5 0s 31ms/step -
accuracy: 0.5547 - loss: 1.2630
Epoch 514: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.5456 - loss: 1.2613 - val_accuracy: 0.7065 - val_loss: 0.7947
Epoch 515/1000
1/5 0s 34ms/step -
accuracy: 0.5078 - loss: 1.2370
Epoch 515: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5555 - loss: 1.1976 - val_accuracy: 0.7114 - val_loss: 0.7927
Epoch 516/1000
1/5 0s 33ms/step -
accuracy: 0.6406 - loss: 1.0980
Epoch 516: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5824 - loss: 1.1848 - val_accuracy: 0.7164 - val_loss: 0.7940
Epoch 517/1000
1/5 0s 39ms/step -
accuracy: 0.5938 - loss: 1.1731
Epoch 517: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5666 - loss: 1.2086 - val_accuracy: 0.7015 - val_loss: 0.7957
Epoch 518/1000
1/5 0s 35ms/step -
accuracy: 0.5859 - loss: 1.2728
Epoch 518: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5798 - loss: 1.2264 - val_accuracy: 0.6866 - val_loss: 0.8005
Epoch 519/1000
1/5 0s 34ms/step -
accuracy: 0.5859 - loss: 1.3317
Epoch 519: saving model to keypoint_classifier.keras
5/5 0s 36ms/step -
accuracy: 0.5702 - loss: 1.2770 - val_accuracy: 0.7264 - val_loss: 0.8000
Epoch 520/1000
1/5 0s 41ms/step -
accuracy: 0.5156 - loss: 1.2903
Epoch 520: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5593 - loss: 1.2425 - val_accuracy: 0.7114 - val_loss: 0.7992
Epoch 521/1000
1/5 0s 34ms/step -
accuracy: 0.5234 - loss: 1.2339

Epoch 521: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5389 - loss: 1.2365 - val_accuracy: 0.7015 - val_loss: 0.7928
Epoch 522/1000
1/5 0s 30ms/step -
accuracy: 0.5234 - loss: 1.2375
Epoch 522: saving model to keypoint_classifier.keras
5/5 0s 31ms/step -
accuracy: 0.5524 - loss: 1.2433 - val_accuracy: 0.7114 - val_loss: 0.7893
Epoch 523/1000
1/5 0s 32ms/step -
accuracy: 0.4922 - loss: 1.2463
Epoch 523: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5173 - loss: 1.2538 - val_accuracy: 0.7164 - val_loss: 0.7880
Epoch 524/1000
1/5 0s 34ms/step -
accuracy: 0.5391 - loss: 1.3257
Epoch 524: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5678 - loss: 1.2055 - val_accuracy: 0.7164 - val_loss: 0.7910
Epoch 525/1000
1/5 0s 44ms/step -
accuracy: 0.5859 - loss: 1.2517
Epoch 525: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5573 - loss: 1.2630 - val_accuracy: 0.7164 - val_loss: 0.7973
Epoch 526/1000
1/5 0s 31ms/step -
accuracy: 0.4844 - loss: 1.3465
Epoch 526: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5311 - loss: 1.2986 - val_accuracy: 0.7363 - val_loss: 0.8015
Epoch 527/1000
1/5 0s 37ms/step -
accuracy: 0.5312 - loss: 1.3069
Epoch 527: saving model to keypoint_classifier.keras
5/5 0s 30ms/step -
accuracy: 0.5453 - loss: 1.2807 - val_accuracy: 0.7114 - val_loss: 0.8000
Epoch 528/1000
1/5 0s 34ms/step -
accuracy: 0.5859 - loss: 1.1494
Epoch 528: saving model to keypoint_classifier.keras
5/5 0s 41ms/step -
accuracy: 0.5707 - loss: 1.2301 - val_accuracy: 0.7164 - val_loss: 0.7974
Epoch 529/1000
1/5 0s 37ms/step -
accuracy: 0.5547 - loss: 1.2129

Epoch 529: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5750 - loss: 1.2062 - val_accuracy: 0.7114 - val_loss: 0.7956
Epoch 530/1000
1/5 0s 33ms/step -
accuracy: 0.6250 - loss: 1.2035
Epoch 530: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5856 - loss: 1.1879 - val_accuracy: 0.7164 - val_loss: 0.7969
Epoch 531/1000
1/5 0s 38ms/step -
accuracy: 0.5156 - loss: 1.2646
Epoch 531: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5475 - loss: 1.2477 - val_accuracy: 0.7214 - val_loss: 0.7973
Epoch 532/1000
1/5 0s 41ms/step -
accuracy: 0.5625 - loss: 1.1799
Epoch 532: saving model to keypoint_classifier.keras
5/5 0s 26ms/step -
accuracy: 0.5904 - loss: 1.1724 - val_accuracy: 0.7264 - val_loss: 0.7935
Epoch 533/1000
1/5 0s 48ms/step -
accuracy: 0.5938 - loss: 1.2483
Epoch 533: saving model to keypoint_classifier.keras
5/5 0s 29ms/step -
accuracy: 0.5882 - loss: 1.2665 - val_accuracy: 0.7214 - val_loss: 0.7959
Epoch 534/1000
1/5 0s 47ms/step -
accuracy: 0.6484 - loss: 1.1345
Epoch 534: saving model to keypoint_classifier.keras
5/5 0s 28ms/step -
accuracy: 0.6076 - loss: 1.1792 - val_accuracy: 0.7164 - val_loss: 0.7968
Epoch 535/1000
1/5 0s 35ms/step -
accuracy: 0.4766 - loss: 1.3021
Epoch 535: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5449 - loss: 1.2320 - val_accuracy: 0.7164 - val_loss: 0.7940
Epoch 536/1000
1/5 0s 30ms/step -
accuracy: 0.6016 - loss: 1.1115
Epoch 536: saving model to keypoint_classifier.keras
5/5 0s 27ms/step -
accuracy: 0.5880 - loss: 1.1674 - val_accuracy: 0.7214 - val_loss: 0.7889
Epoch 537/1000
1/5 0s 36ms/step -
accuracy: 0.6016 - loss: 1.1894

```

Epoch 537: saving model to keypoint_classifier.keras
5/5          0s 42ms/step -
accuracy: 0.5884 - loss: 1.1700 - val_accuracy: 0.7313 - val_loss: 0.7891
Epoch 538/1000
1/5          0s 45ms/step -
accuracy: 0.5625 - loss: 1.1886
Epoch 538: saving model to keypoint_classifier.keras
5/5          0s 27ms/step -
accuracy: 0.5649 - loss: 1.2014 - val_accuracy: 0.7463 - val_loss: 0.7926
Epoch 539/1000
1/5          0s 38ms/step -
accuracy: 0.5469 - loss: 1.2007
Epoch 539: saving model to keypoint_classifier.keras
5/5          0s 29ms/step -
accuracy: 0.5470 - loss: 1.2389 - val_accuracy: 0.7214 - val_loss: 0.7914
Epoch 540/1000
1/5          0s 32ms/step -
accuracy: 0.5469 - loss: 1.2479
Epoch 540: saving model to keypoint_classifier.keras
5/5          0s 31ms/step -
accuracy: 0.5674 - loss: 1.2163 - val_accuracy: 0.7114 - val_loss: 0.7895
Epoch 541/1000
1/5          0s 35ms/step -
accuracy: 0.5000 - loss: 1.3974
Epoch 541: saving model to keypoint_classifier.keras
5/5          0s 31ms/step -
accuracy: 0.5167 - loss: 1.3149 - val_accuracy: 0.7015 - val_loss: 0.7922
Epoch 542/1000
1/5          0s 43ms/step -
accuracy: 0.6094 - loss: 1.1580
Epoch 542: saving model to keypoint_classifier.keras
5/5          0s 33ms/step -
accuracy: 0.5683 - loss: 1.2194 - val_accuracy: 0.7015 - val_loss: 0.7949
Epoch 543/1000
1/5          0s 40ms/step -
accuracy: 0.4844 - loss: 1.3261
Epoch 543: saving model to keypoint_classifier.keras
5/5          0s 38ms/step -
accuracy: 0.5510 - loss: 1.2356 - val_accuracy: 0.7015 - val_loss: 0.7901
Epoch 543: early stopping

```

```
[34]: val_loss, val_acc = model.evaluate(X_test, y_test, batch_size=128)
```

```

2/2          0s 27ms/step -
accuracy: 0.7151 - loss: 0.7735

```

```
[35]: model = tf.keras.models.load_model(model_save_path)
```

```
[36]: predict_result = model.predict(np.array([X_test[0]]))
print(np.squeeze(predict_result))
print(np.argmax(np.squeeze(predict_result)))
```

```
1/1          0s 64ms/step
[7.28776330e-30  1.04502730e-37  1.44501622e-22  6.49147321e-37
 3.75445743e-28  4.43530434e-36  6.35864916e-10  5.13876808e-09
 1.30228614e-28  2.97415095e-33  7.91637976e-29  1.90886909e-34
 2.72557046e-19  1.31418055e-17  2.16530628e-23  1.20146945e-01
 8.79853070e-01  1.24421359e-23  9.87651329e-22  4.56829602e-15
 2.45617559e-26  2.80492847e-37  2.55112626e-34  6.08814495e-33
 0.00000000e+00  2.64984791e-27]
```

```
16
```

```
[37]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report

def print_confusion_matrix(y_true, y_pred, report=True):
    labels = sorted(list(set(y_true)))
    cmx_data = confusion_matrix(y_true, y_pred, labels=labels)

    df_cmx = pd.DataFrame(cmx_data, index=labels, columns=labels)

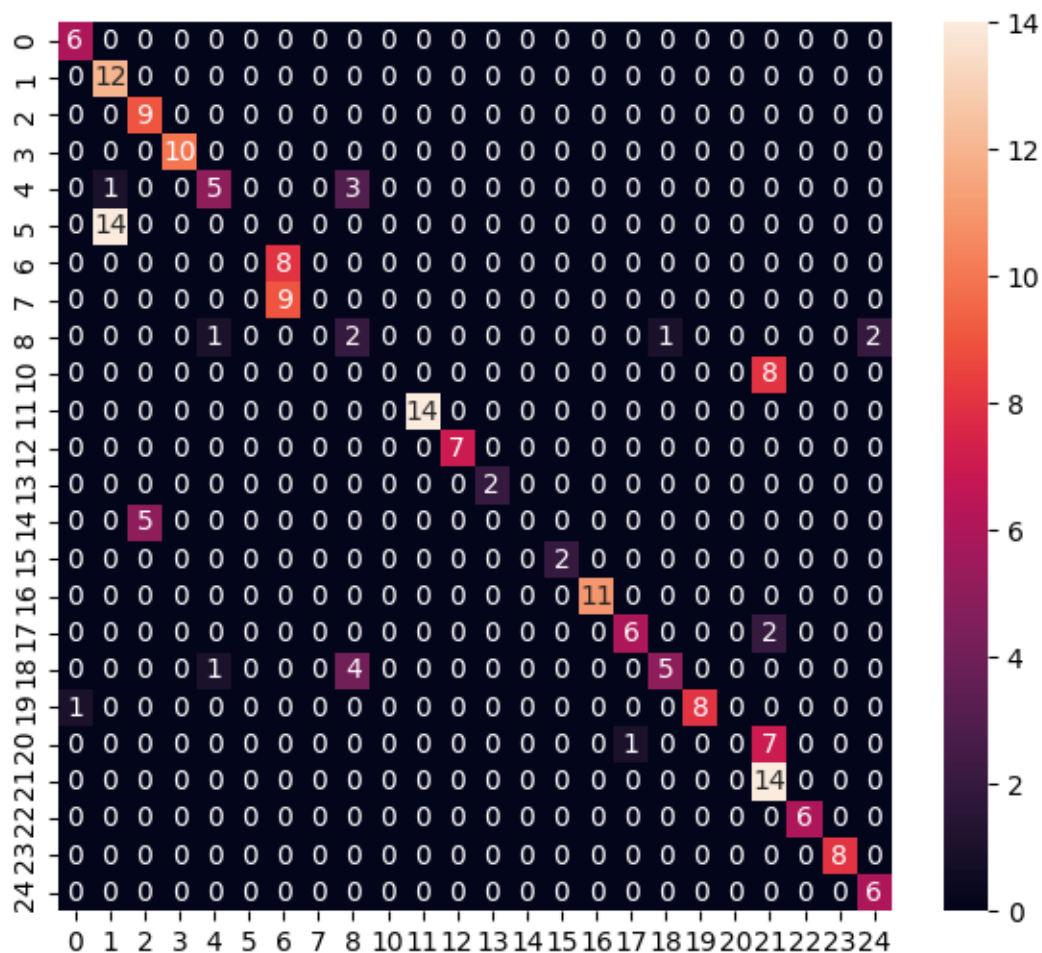
    fig, ax = plt.subplots(figsize=(7, 6))
    sns.heatmap(df_cmx, annot=True, fmt='g', square=False)
    ax.set_ylim(len(set(y_true)), 0)
    plt.show()

    if report:
        print('Classification Report')
        print(classification_report(y_test, y_pred))

Y_pred = model.predict(X_test)
y_pred = np.argmax(Y_pred, axis=1)

print_confusion_matrix(y_test, y_pred)
```

```
7/7          0s 4ms/step
```



Classification Report

	precision	recall	f1-score	support
0	0.86	1.00	0.92	6
1	0.44	1.00	0.62	12
2	0.64	1.00	0.78	9
3	1.00	1.00	1.00	10
4	0.71	0.56	0.62	9
5	0.00	0.00	0.00	14
6	0.47	1.00	0.64	8
7	0.00	0.00	0.00	9
8	0.22	0.33	0.27	6
10	0.00	0.00	0.00	8
11	1.00	1.00	1.00	14
12	1.00	1.00	1.00	7
13	1.00	1.00	1.00	2
14	0.00	0.00	0.00	5

15	1.00	1.00	1.00	2
16	1.00	1.00	1.00	11
17	0.86	0.75	0.80	8
18	0.83	0.50	0.62	10
19	1.00	0.89	0.94	9
20	0.00	0.00	0.00	8
21	0.45	1.00	0.62	14
22	1.00	1.00	1.00	6
23	1.00	1.00	1.00	8
24	0.75	1.00	0.86	6
accuracy			0.70	201
macro avg	0.64	0.71	0.65	201
weighted avg	0.61	0.70	0.63	201

```
c:\Users\Reetam\AppData\Local\Programs\Python\Python310\lib\site-
packages\sklearn\metrics\_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
c:\Users\Reetam\AppData\Local\Programs\Python\Python310\lib\site-
packages\sklearn\metrics\_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
c:\Users\Reetam\AppData\Local\Programs\Python\Python310\lib\site-
packages\sklearn\metrics\_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
[38]: model.save(model_save_path, include_optimizer=False)
```

```
[39]: # tf_lite_save_path = 'model/keypoint_classifier/keypoint_classifier.tflite'
tflite_save_path = 'keypoint_classifier.tflite'

converter = tf.lite.TFLiteConverter.from_keras_model(model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
tflite_quantized_model = converter.convert()

open(tflite_save_path, 'wb').write(tflite_quantized_model)
```

```
INFO:tensorflow:Assets written to:
C:\Users\Reetam\AppData\Local\Temp\tmpxxm68vox\assets
```

```
INFO:tensorflow:Assets written to:
C:\Users\Reetam\AppData\Local\Temp\tmpxxm68vox\assets
```

```
Saved artifact at 'C:\Users\Reetam\AppData\Local\Temp\tmpxxm68vox'. The
```

following endpoints are available:

```
* Endpoint 'serve'
  args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 42), dtype=tf.float32,
name='input_layer_1')
Output Type:
  TensorSpec(shape=(None, 26), dtype=tf.float32, name=None)
Captures:
  2263443959376: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2261344252000: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2263443961664: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2261344242496: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2261344245488: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2261344392240: TensorSpec(shape=(), dtype=tf.resource, name=None)
```

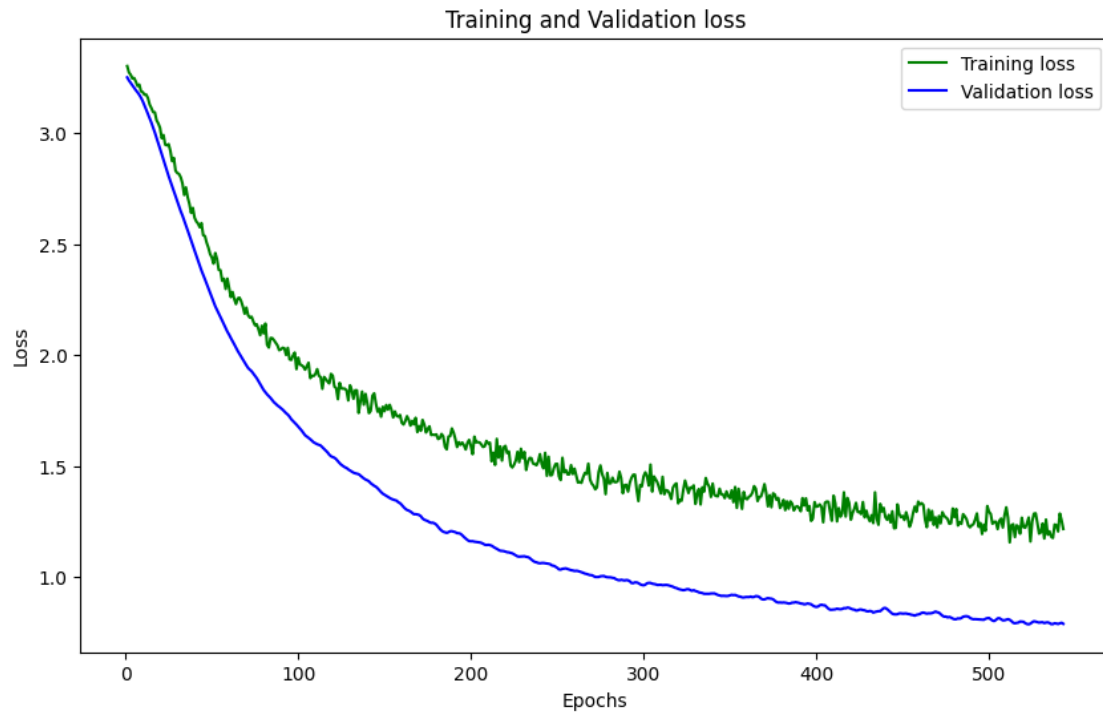
[39]: 7524

```
[40]: interpreter = tf.lite.Interpreter(model_path=tflite_save_path)
      interpreter.allocate_tensors()
```

```
c:\Users\Reetam\AppData\Local\Programs\Python\Python310\lib\site-
packages\tensorflow\lite\python\interpreter.py:457: UserWarning:      Warning:
tf.lite.Interpreter is deprecated and is scheduled for deletion in
    TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
    See the [migration guide](https://ai.google.dev/edge/litert/migration)
    for details.
```

```
warnings.warn(_INTERPRETER_DELETION_WARNING)
```

```
[41]: loss_train = history.history['loss']
      loss_val = history.history['val_loss']
      epochs = range(1, len(loss_train) + 1)
      plt.figure(figsize=(10, 6))
      plt.plot(epochs, loss_train, 'g', label='Training loss')
      plt.plot(epochs, loss_val, 'b', label='Validation loss')
      plt.title('Training and Validation loss')
      plt.xlabel('Epochs')
      plt.ylabel('Loss')
      plt.legend()
      plt.show()
```



```
[42]: acc_train = history.history['accuracy']
acc_val = history.history['val_accuracy']
epochs = range(1, len(acc_train) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, acc_train, 'g', label='Training accuracy')
plt.plot(epochs, acc_val, 'b', label='Validation accuracy')
plt.title('Training and Validation accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()
```

